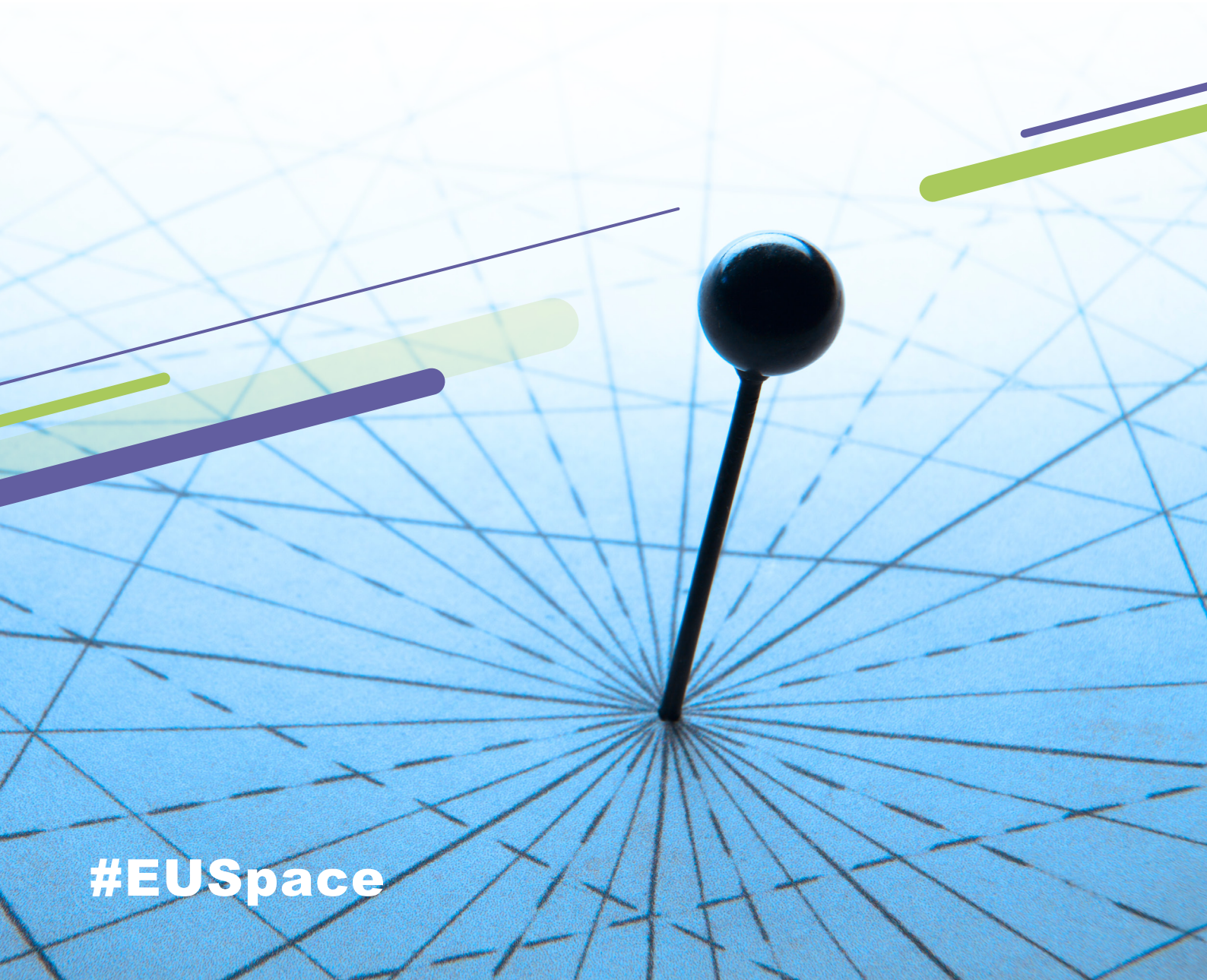
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GALILEO HIGH ACCURACY SERVICE REFERENCE USER ALGORITHM (**HAS UA**) - TECHNICAL NOTE



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1. INTRODUCTION

1.1. Scope, objectives and target audience

The Galileo High Accuracy Service (HAS) is an open access [1] service broadcasting Precise Point Positioning (PPP) corrections through the data component (E6-B) of the Galileo E6-B/C signal. The HAS corrections are transmitted at 448 bits per second by each transmitting Galileo satellite [2]. Furthermore, HAS corrections are also distributed through a terrestrial dissemination channel, enabling users with no E6-B reception/tracking capabilities to access the service through the internet [3].

HAS will be delivered in two phases, indicated as Initial Service and Full Service [4]. Since the Initial Service declaration, precise orbits, clock and biases for Galileo and GPS constellations are provided. After the declaration of the Full Service, further features will also be broadcast in addition to the above corrections.

This document presents the Galileo HAS Reference User Algorithm (UA) – referred to as HAS-UA – including the algorithm assumptions and mathematical background to facilitate the algorithm use and implementation. It is a real-time PPP algorithm that exploits the HAS corrections to generate a Position, Velocity, Time (PVT) solution and that allows achieving the target HAS Minimum Performance Levels (MPLs) and target performance specified in the HAS Service Definition Document (SDD) [4].

Note$_1$: *This document should be considered as a complement to the Galileo HAS Interface Control Document (ICD) [2].*

Note$_2$: *This document is not tailored to any specific application but it is intended to assist each HAS user interested in implementing or using the HAS-UA to obtain a HAS PVT. Alternative algorithms, based on the same principles, could also be used to meet the Galileo HAS performance.*

1.2. Structure of the document

This document is structured as follows:

- Chapter 1 provides the overall scope, objectives and target audience;
- Chapter 2 illustrates the HAS-UA and its main functionalities. Moreover, it details its external and internal interfaces. Finally, it provides the HAS workflow;
- Chapter 3 provides the HAS-UA theoretical background. Additional aspects are covered in the appendices;
- Appendix A: provides details on the computation of the antenna phase centre variation (PCV);
- Appendix B: provides the notation definition for the equations included in this document;
- Appendix C: provides the rotation matrices for the application of the phase centre offsets from the ANTEX files;
- Appendix D: explains how to compute the Sun position;
- Appendix E: provides details on the phase wind-up effect computation;
- Appendix F: specifies an example of numerical application of the user algorithm.

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2. HAS-UA FUNCTIONALITIES AND INTERFACES

This section provides a general overview of the HAS-UA functional blocks, describing its main stages, interfaces and working modes. Finally, it introduces the different steps within the HAS-UA workflow.

2.1. HAS-UA overview

The HAS-UA is composed of two main blocks, i.e. the Pre-Processing (PP) and the NAV estimation blocks, whose main functionalities are reported in the following lines, while a more detailed description is given in Sections 2.4.1 and 2.4.2, respectively.

Pre-Processing (PP): this block is in charge of performing the HAS-UA input data pre-processing. This task includes the data quality check, cycle slip (CS) and clock jump (CJ) detection and the estimation of a single epoch (SE) PVT solution (with the application of HAS corrections) from undifferenced (i.e. no combination in time or with measurements from a pivot satellite) ionosphere-free combined pseudorange and uncombined Doppler measurements. The SE PVT solution is used for the initialization of the NAV estimation when needed, the determination of the NAV process noise and the computation of the signal travel time.

NAV estimation: this block is in charge of estimating the NAV solution by exploiting the pre-processed measurements and the data provided by the PP block. The NAV solution is estimated using a PPP algorithm based on an Extended Kalman Filter (EKF) processing single difference (SD[1]) of uncombined (i.e. no combination of measurements from different signals) measurements. The proposed EKF exploits an adaptive approach to tune its process noise covariance matrix according to the estimated user speed range. The approach makes the algorithm flexible to the user dynamic changes and suitable for a wide range of velocities (up to 412 m/s). Finally, the HAS-UA NAV provides a float–ambiguity-based positioning solution.

The HAS-UA requires as input the GNSS measurements, the Broadcast (BC) navigation data, the HAS corrections, the receiver and satellite Phase Centre Offsets (PCOs), and the Phase Centre Variations (PCVs). The PCOs and PCVs are used to refer the HAS corrections to the specific target signal of the uncombined single-differenced measurements used by the HAS-UA. Optionally, also Earth Rotation Parameters (ERPs) can be provided as input, although these parameters are not required for using the HAS UA in the initial phase.

The HAS corrections can be retrieved from the E6-B navigation message (C/NAV pages) after the message demodulation, as illustrated in Figure 1, where the scheme of a generic E6-B receiver implementing the HAS-UA is represented. Alternatively, a Networked Transport of RTCM via Internet Protocol (NTRIP) client can receive the HAS corrections via a terrestrial link established through the Ethernet or Wi-Fi internet-connection interface connected to the NTRIP caster. Please refer to the respective SIS and IDD ICDs for the implementation details.

[1] In the rest of the document, SD will refer to this satellite-satellite type of differences if not otherwise specified.

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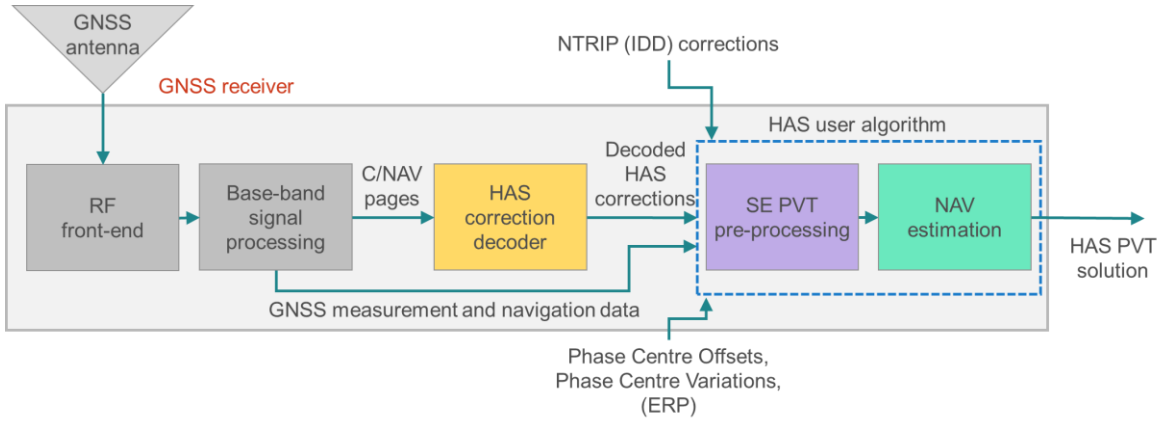


Figure 1: Scheme of an E6-B capable GNSS receiver implementing the HAS-UA.

2.2. HAS-UA interfaces

The HAS-UA is composed of two external interfaces, represented in grey in the Figure **2** and detailed in this section.

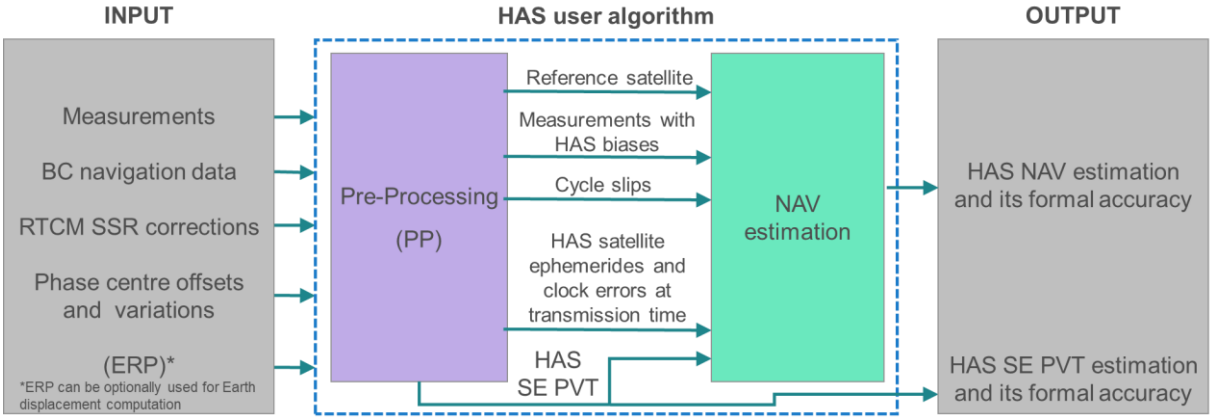


Figure 2: Galileo HAS User Algorithm and its internal and external interfaces.

A detailed list of the input and output data provided by the HAS-UA external interfaces is reported in Table 1.

Table 1: HAS-UA external interfaces output.

External Interface	External Interface Output
INPUT	• Time tagged Galileo and GPS carrier phase, pseudorange, and Doppler measurements, Carrier-to-Noise-Density Ratio (C/N0), phase tracking status and loss of lock indicators. • Galileo and GPS BC navigation data indicated in the HAS message as specified by the HAS ICDs. • HAS orbit and clock corrections, code and phase biases. • Satellite and receiver PCOs and PCVs. • (Optional) ERP, polar motion, ΔUT1.

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External Interface	External Interface Output
OUTPUT	• NAV position solution referred to the Galileo Terrestrial Reference System (GTRF). • NAV velocity solution referred to the GTRF. • PP timing information referred to the Galileo System Time (GST). • Formal accuracy (sigma value) of the position solution. • Formal accuracy (sigma value) of the velocity solution. • Formal accuracy (sigma value) of the time solution.

Together with the external interfaces, there is also an internal interface between the two main UA blocks, PP and NAV, as shown in Figure 2. This unidirectional interface inputs the following data, provided by the PP, to the NAV estimation block:

- HAS corrected satellite positions and clock offsets at transmission time,
- GNSS measurements with HAS biases applied,
- satellite elevations,
- reference satellite,
- cycle slips,
- PP solution, (indicated as HAS SE PVT in Figure 2),
- satellite PCOs and PCVs.

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2.3. HAS-UA working modes

The HAS-UA can work in either double or triple frequency modes supporting both Galileo single constellation and Galileo plus GPS dual constellation mode. Specifically, the HAS-UA supports the following working modes.

- Dual Frequency modes:
  - Galileo-only: E1C-E5aQ/E5bQ/E6B[2].
  - Galileo and GPS: E1C-E5aQ/E5bQ/E6B[2] + L1CA-L2C.
- Triple Frequency modes:
  - Galileo-only: E1C-E5aQ/E5bQ-E6B[2].
  - Galileo and GPS: E1C-E5aQ/E5bQ-E6B[2] + L1CA-L2C.

The algorithm processes healthy/marginal SIS satellites [1] as defined in the Galileo HAS Service Definition Document (SDD) [4] (see Figure 6 of the SDD for the usability of satellites in Galileo HAS).

2.4. HAS-UA workflow

In this section, the workflow of the PP and NAV estimation blocks is introduced. Detailed schemes of the PP and NAV blocks, previously introduced in Section 2.1, are shown in Figure 3 and Figure 4, respectively. The HAS-UA sequential steps are reported in Table 2.

Table 2: Sequential steps for the implementation of the HAS-UA.

Block	Execution	Step.
	FOR every epoch *k* DO	
Pre-Processing	Determine the available measurements of the observed GNSS satellites	1
	Determine the HAS-correction available for the observed GNSS satellites	
	Apply the available and valid HAS biases to the GNSS measurements	
	Determine the broadcast data to use and compute the satellite positions, velocities and clock errors from broadcast message	2
	Apply available and valid HAS satellite position and clock error corrections to the computed satellite positions, velocities and clock errors	
	Estimate absolute single epoch receiver position, velocity and clock offset and drift using pseudorange and Doppler measurements by WLSE	3
	Perform outlier detection, identification and adaptation	4
	Detect and correct cycle slips in carrier phase measurements	5
	Select reference satellite as the satellite of highest elevation among all satellites with sufficient quality including HAS corrections and non-eccentric orbits	6
	IF *k* == 1 OR state vector needs total initialisation	7
	Initialize state/measurement vectors and their covariance matrices, with receiver-related states initialised from the PP. Set all satellites as rising.	
	ELSE	8

[2] E6C can also be used if not encrypted.

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Block	Execution	Step.
	Perform linear prediction of state estimates and their covariance matrix	
	END	
	Compute satellite-satellite single differences (SD) of corrected measurements in input to EKF	9
	Compute design matrix	10
	IF measurements of new satellites or new frequencies are available or phase discontinuities occur	11-12
	Initialize ionospheric delays and ambiguities	
	END	
	Compute EKF adaptive process noise	13
	Measurement update	14
	Detect outliers from estimated residuals	15
	Apply Displacement Correction for ITRF compatible PPP solution. (optional)	16
	END	

2.4.1. PP workflow

In Table 3 and Table 4 the main steps necessary for the implementation of the two HAS-UA blocks are reported, along with the corresponding document section where each step is described in detail.

Table 3: PP steps.

Step	Step denomination	Description	Section
1	Data Collection and Synchronization	Initial preparation of the input data	3.2.1
	Computation of HAS corrected GNSS measurements	It applies the HAS biases to the pseudorange and phase measurements.	3.2.2
2	Computation of HAS satellite ephemerides and clock errors from BC and HAS data	It provides the HAS satellite ephemeris and clocks errors from the BC and HAS data as described by the Galileo OS [1], HAS [2][3] and GPS [5] ICDs.	3.2.2 3.2.3
3	Determination of absolute receiver position, velocity, clock offset and drift with iterative weighted least squares estimation (WLSE) [17] in the SE PVT solution	It provides the receiver position, velocity and clock offset and drift estimated from the satellite ephemeris and clock errors and from the pseudorange and Doppler measurements obtained in the previous steps. It also provides the final satellite ephemerides and clock errors at transmission time. The estimation is performed by applying an iterative WLSE.	3.2.4 3.2.5

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Step	Step denomination	Description	Section
		The solution is then used for the initialization of the NAV estimator. The SE PVT estimated time solution is also provided as output to the user. The velocity solution is also exploited to determine the tuning of the EKF process noise used for the NAV estimation, as described in Section 3.3.7.	
4	Outlier detection, identification and adaptation	It performs a test statistic check on the SE PVT residuals. If measurements are rejected the SE PVT solution is adapted without need to re-estimate the solution.	3.2.6
5	Cycle slip detection and correction	It detects and corrects CSs of pre-corrected SD carrier phase measurements. The CS determination is based on the joint usage of two linear signal combinations and the receiver loss of lock indicator (when available). The CS tests are repeated after CS reparation and, in case of CS detection and correction failure, the corresponding ionosphere and ambiguity parameters in the NAV estimation state vector and their process noise must be reinitialised.	3.2.7
6	Reference satellite selection	It indicates a reference satellite for each constellation. The selected satellite is used as the pivot satellite for the calculation of the SD of the measurements.	3.2.8

2.4.2. NAV estimation workflow

The NAV estimation is implemented by using the PP output and applying the steps reported in Table 4 and illustrated in Figure 4. In Table 4, along with the steps description and the reference to the associated sections, the execution phase is also reported. In particular, it is indicated that step 7 is performed during the initialization phase while the steps from 8 to 16 are included in the iteration loop and executed at each epoch as also detailed in Section 2.4 and 3.3.

Table 4: NAV estimation steps.

Step	Step denomination	Description	Section
7	Initialization of the receiver parameters and their covariance matrix in the NAV state vector (if the state vector is not initialised yet or its reset is required)	It initializes the receiver parameters and their covariance matrix in the NAV state vector from the PP SE PVT estimation. All the satellites are set as rising if the initialisation is triggered.	3.3.1

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Step	Step denomination	Description	Section
8	Linear prediction of state parameters and their covariance matrix (if receiver parameters are not initialised)	It performs a linear prediction of the estimated receiver position, velocity, tropospheric zenith, ionospheric slant delays and phase ambiguities. This step is an alternative to Step 7.	3.3.2
9	Computation of HAS corrected satellite-satellite SD measurements	It provides satellite-satellite SD of carrier phase, pseudorange and Doppler measurements and their variances, formed using the pivot satellite obtained at step 6, corrected by the respective SD satellite errors. The applied satellite errors include the projections of the HAS-corrected satellite positions into the satellite-receiver line of sight (LOS), the HAS-corrected satellite clock errors and the phase and code bias SD corrections. The measurement noise covariance matrix is obtained by using a satellite elevation-dependent mapping function of the zenithal GNSS measurement accuracy combined with the correction accuracies.	3.3.3
10	Design matrix determination	It provides the design matrix relating the vector of the unknown parameters and the HAS corrected SD carrier phase, pseudorange and Doppler measurements.	3.3.3
11	Initialization of ionosphere and ambiguity states of newly tracked (rising) satellites	It initializes the ionospheric slant delays and ambiguities of newly tracked rising satellites using the measurements of the newly tracked satellites and the receiver parameters from the latest estimation of the state vector.	3.3.5
12	Ambiguities and ionosphere slant delay reinitialization due to phase discontinuity	It reinitialises the ambiguity and the ionosphere states for those signals marked as affected by unrepaired CS.	3.3.6
13	Computation of EKF process noise covariance matrix	It determines the EKF process noise covariance matrix using an adaptive setting depending on the user velocity.	3.3.7
14	Measurement update	It performs the update of the predicted states with the actual measurements. First, the EKF gains are determined. Then, the state parameters and their covariance matrix are	3.3.8

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Step	Step denomination	Description	Section
		updated using the measurement residuals of the state prediction obtained at Step 9.	
15	Outlier detection, identification and adaptation	It performs a test statistic check on the NAV innovation residuals. If measurements are rejected the NAV solution is adapted without need to re-estimate the solution as described in Section 3.2.6.	3.3.9
16	Applications of site displacement corrections (optional)	This block determines the site displacement corrections for the Earth tides, polar motion and ocean loading and applies these corrections to the receiver position coordinates. This step is optional for the user because no ERP are provided by the service in its initial service. To perform this step, ERP data should be independently procured. Moreover, this correction is useful to compute the marker position of static receivers. For kinematic receivers it is not a necessary correction. See section 5.7 of [8].	3.3.10

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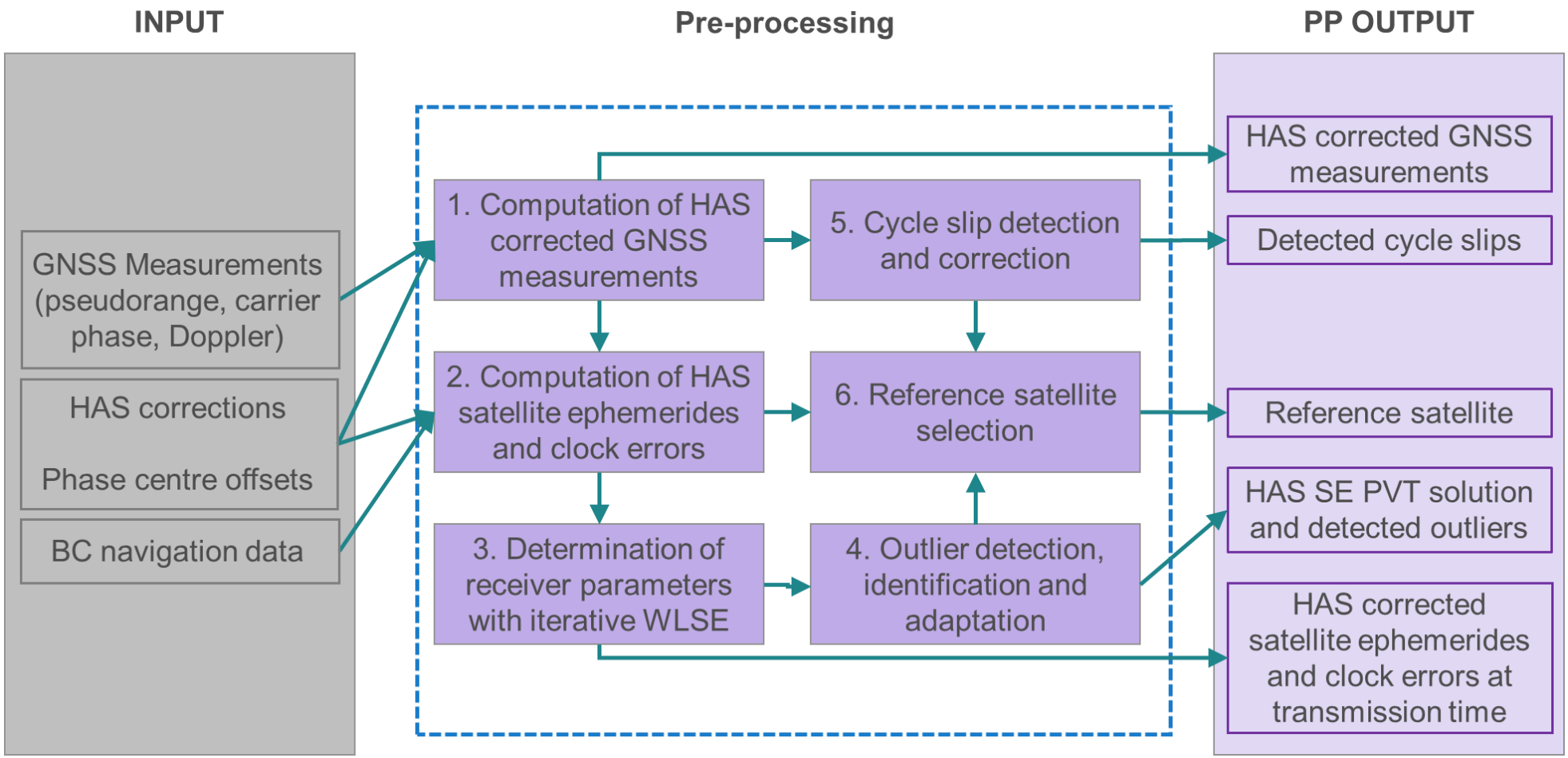


Figure 3: Diagram of the HAS-UA PP

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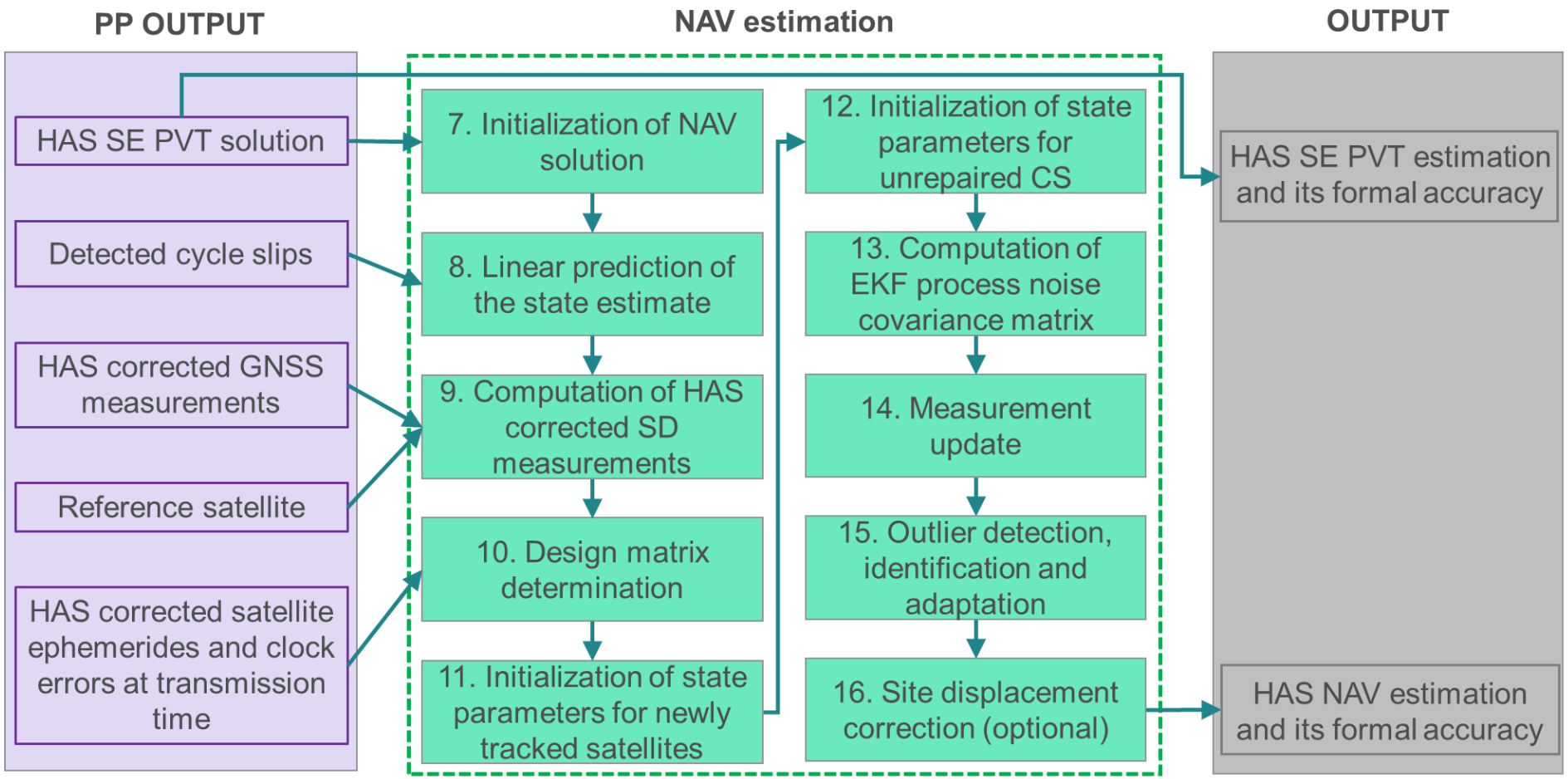


Figure 4: Diagram of the HAS-UA PVT Estimation

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3. HAS-UA DESCRIPTION

This section details the different HAS-UA steps, for both PP and NAV estimations blocks, introduced in subsections 2.4.1 and 2.4.2 respectively.

As mentioned in subsection 2.1, the PP block is in charge of pre-processing the input measurements and providing all the input data necessary for the estimation of the NAV solution. This task includes the application of the HAS corrections to the raw measurements, the detection and removal of measurement outliers and the estimate of the standard PVT solution used to initialize the HAS PPP.

3.1. Model for carrier phase, pseudorange and Doppler measurements

In the HAS-UA, carrier phase and pseudorange measurements are corrected by using HAS corrections and biases to obtain the NAV solution. For a more detailed explanation of the different terms composing these measurements, the interested reader can refer to [6] and [7]. In the following, a variable indicated with an underbar (e.g. $\underline{y}$) is considered to be a random variable.

The carrier phase measurement, in unit of length, can be modelled as

$$\begin{aligned}\underline{\Phi}^s_{u,m}(t') &= \rho^s_u(t, t-\tau^s_u) + dt^\Sigma_u(t) - dt^s(t-\tau^s_u) + T^s_u(t) - \mu_m I^s_u(t) + \lambda_m \beta_{u,m}(t) - \lambda_m \beta^s_m(t) \\ &+\zeta^s_{\mathrm{PCO},u,m}(t) + \zeta^s_{\mathrm{PCV},u,m}(t) + \lambda_m W^s_{\mathrm{u,m}}(t) + \delta t^s_{\mathrm{Sha},u}(t) + \Delta\rho^s_{u,m}(t) + \lambda_m M^s_{u,m}(t_0) + \underline{\varepsilon}^s_{u,m}(t)\end{aligned} \quad (1)$$

where

- $s \in \{1, \ldots, S\}$ indicates the tracked satellite;
- $m \in \{1, \ldots, M\}$ indicates the combination of system-carrier-modulation signal (one independent index for each combination);
- u indicates the user receiver;
- $\Sigma \in \{E, G\}$ indicates the user GNSS constellation, E for Galileo and G for GPS;
- $t' = t_u(t) = t + \frac{dt^\Sigma_u(t)}{c}$ is the measurement epoch in the receiver time frame;
- c is the speed of the light as indicated in the OS Galileo and IS GPS ICDs;
- t is the measurement epoch in the reference system time frame, i.e. the Galileo System Time (GST) for Galileo measurement and the GPS time (GPST) for GPS measurements, namely the selected epoch at which to process the observations;
- $dt^\Sigma_u(t) = dt^\Sigma_u$ is the clock error in unit of length for the user u receiver w.r.t. the reference time of the GNSS constellation Σ;
- $dt^s(t) = dt^s$ is the clock error (including the relativity effect due to the orbit eccentricity) in unit of length for the satellite s transmitter w.r.t. the reference time of its GNSS constellation $\Sigma(s)$;
- λ_m is the wavelength of the carrier signal m;
- $\rho^s_u(t, t-\tau^s_u) = \rho^s_u = \|\mathbf{r}_u(t) - \mathbf{r}^s(t-\tau^s_u)\|$ indicates the geometrical distance between satellite s centre of mass (COM) at transmission time (computed as indicated in section 3.2.4) and user u antenna reference point (ARP) at reception time, both referred to the Galileo HAS reference frame, GTRF, (defined in Section 7.2 of [2] for SIS and Section 3.3.2 of [3] for IDD) at reception time;

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- $\mathbf{r}_u(t) = \mathbf{r}_u$ indicates the user u position referred to the ARP in the HAS reference frame at the time of signal reception;
- τ_u^s is the signal travel time, i.e. the time difference between the signal transmission and reception times as if the signal was traveling in a vacuum;
- $\mathbf{r}^s(t-\tau_u^s) = \mathbf{r}^s$ indicates the satellite location referred to the satellite COM at the time of signal transmission in the HAS reference frame at reception time (see Section 3.2.2 for details on its computation);
- $\mathbf{e}_u^s(t, t-\tau_u^s) = \mathbf{e}_u^s = \frac{\mathbf{r}_u - \mathbf{r}^s}{\|\mathbf{r}_u - \mathbf{r}^s\|}$ is the normalized direction vector pointing from the satellite s to the user u receiver location;
- $\zeta^s_{\mathrm{PCO},u,m}(t) = \zeta^s_{\mathrm{PCO},u,m} = \mathbf{e}_u^{s\,T}(t, t-\tau_u^s)\left(\mathbf{A}_\mathrm{u}\delta\mathbf{r}_{\mathrm{PCO},u,m}(t) - \mathbf{A}^\mathrm{s}\delta\mathbf{r}^\mathrm{s}_{\mathrm{PCO},m}(t-\tau_u^s)\right)$ is the projection on the line of sight (LOS) $\mathbf{e}_u^s(t, t-\tau_u^s)$ between the user u receiver and the satellite s of their antenna PCO for the signal m, i.e. the m antenna phase centre (APC_m) w.r.t. the user u ARP or the satellite s COM, rotated in the GTRF frame through the rotation matrices $\mathbf{A}^\mathrm{s}$ and $\mathbf{A}_\mathrm{u}$ (see Appendix C for details);
- $\zeta^s_{\mathrm{PCV,u,m}}(t) = \zeta^s_{\mathrm{PCV,u,m}} = \zeta^s_{\mathrm{PCV,m}} + \zeta_{\mathrm{PCV,u,m}}$ is the user u and satellite s antenna PCV for the signal m (see Appendix A for details);
- $T_u^s(t) = T_u^s = m_{h,u}^s T_{h,u} + m_{w,u}^s T_{w,u}$ is the total (wet + hydrostatic) tropospheric slant delay between the user u and the satellite s;
- $m^s_{\mathrm{h},u}$ is the hydrostatic troposphere mapping function from zenith to the line of sight between the user u and the satellite s;
- $T_{\mathrm{h},u}$ is the hydrostatic troposphere zenith delay at the user u receiver location;
- $m^s_{\mathrm{w},u}$ is the wet troposphere mapping function from zenith to the line of sight between the user u and the satellite s;
- $T_{\mathrm{w},u}$ is the wet troposphere zenith delay at the user u receiver location;
- $\mu_m = \frac{f_1^2}{f_m^2} = \frac{\lambda_m^2}{\lambda_1^2}$ is the ratio between the ionosphere delay in the L1/E1 reference carrier and the selected carrier m;
- $I_u^s(t) = I_u^s$ is the first order ionosphere slant delay on L1/E1 frequency (1575.42 MHz) between the user u and the satellite s;
- $M_{u,m}^s(t_0) = M_{u,m}^s = \phi_{u,m}(t_0) - \phi_m^s(t_0) + N_{u,m}^s(t_0)$ is the carrier phase ambiguity in unit of cycles for the signal m between the user u and the satellite s;
- $\phi_{u,m}(t_0)$ is the initial phase offset in unit of cycles on user receiver u for signal m since the beginning of signal tracking t_0;
- $\phi_m^s(t_0)$ is the initial phase offset in unit of cycles on satellite s for signal m;
- $N_{u,m}^s(t_0) = N_{u,m}^s$ is the integer carrier phase ambiguity in unit of cycles for the signal m between the user u and the satellite s;
- $\beta_{u,m}(t) = \beta_{u,m}$ is the user receiver phase bias in unit of cycles for the signal m;
- $\beta_m^s(t) = \beta_m^s$ is the satellite phase bias in unit of cycles for the signal m;
- $W^s_{\mathrm{u,m}}(t) = W^s_{\mathrm{u,m}}$ is the satellite phase wind-up error, in unit of cycles, between the user receiver u and the satellite s (see Appendix E for details);
- $\delta t^s_{\mathrm{Sha},u}(t) = \delta t^s_{\mathrm{Sha,u}} = \frac{2\mu}{c}\ln\left(\frac{\|\mathbf{r}^s\| + \|\mathbf{r}_u\| + \rho_u^s}{\|\mathbf{r}^s\| + \|\mathbf{r}_u\| - \rho_u^s}\right)$ is the relativistic correction in unit of length between the user receiver u and the satellite s for the Shapiro signal propagation delay, with μ the Earth gravitational constant;

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- $\Delta\rho^s_{u,m}(t) = \Delta\rho^s_{u,m}$ is the phase multipath error for the signal m from satellite s to the user u;
- $\underline{\varepsilon}^s_{u,m}(t) = \underline{\varepsilon}^s_{u,m} \sim \mathrm{N}(0, \sigma^2_\varepsilon)$ is the phase noise for the signal m from satellite s received by the user u, which is assumed to have a Normal distribution with mean zero and variance σ^2_ε.

Omitting from here on the time dependencies of the parameters as clear extension of the meaning shown here above, the pseudorange measurements can be expressed as follows

$$\underline{P}^s_{u,m} = \rho^s_{u,m} + dt^\Sigma_u - dt^s + T^s_u + \mu_m I^s_u + b_{u,m} - b^s_m + \zeta^s_{\mathrm{PCO,u,m}}(t) + \delta t^s_{\mathrm{Sha}} + \Delta r^s_{u,m} + \underline{\epsilon}^s_{u,m} \tag{2}$$

where

- $b_{u,m}$ is the receiver code bias for the system-carrier-modulation signal m;
- b^s_m is the satellite code bias for the system-carrier-modulation signal m;
- $\Delta r^s_{u,m}$ is the pseudorange multipath error for the signal m from satellite s received by the user u;
- $\underline{\epsilon}^s_{u,m} \sim \mathrm{N}(0, \sigma^2_\epsilon)$ is the receiver pseudorange noise for the signal m from satellite s received by user u.

The Doppler measurements provided by a GNSS receiver can be expressed, in Hz, as

$$\underline{d}^s_{u,m} = -\frac{f_m}{c}\dot{\rho}^s_{u,m} + df_{u,m} - df^s_m + \Delta f^s_{u,m} + \underline{\xi}^s_{u.m} \tag{3}$$

where

- f_m is the nominal central frequency of the signal m;
- $\dot{\rho^s_u} = \dot{\rho}^s_u(t, t - \tau^s_u) = \|\mathbf{v}_u(t) - \mathbf{v}^s(t - \tau^s_u)\|$ indicates the geometrical distance rate between satellite s at transmission time and user u at reception time;
- $\mathbf{v}_u(t) = \mathbf{v}_u$ is the velocity of the user u receiver at reception time;
- $\mathbf{v}^s(t - \tau^s_u) = \mathbf{v}^s$ is the satellite s velocity at transmission time;
- $df_{u,m} = f_m \frac{dt_u}{c}$ is the frequency deviation of the user u clock, expressed in Hz;
- $\dot{dt}_u$ is the derivative of the user u receiver error, i.e. the user clock drift, expressed in m/s;
- $df^s_m = f_m \frac{dt^s}{c}$ is the frequency deviation of the satellite s clock, expressed in Hz;
- $\dot{dt}^s$ is the derivative of the satellite s clock error (including the relativity effect due to the orbit eccentricity), i.e. the satellite clock drift, expressed in m/s;
- $\Delta f^s_{u,m}$ is the Doppler multipath error for the signal m from satellite s received by the user u;
- $\underline{\xi}^s_{u,m} \sim \mathrm{N}(0, \sigma^2_\eta)$ (and later $\underline{\Xi}^s_{u.m} \sim \mathrm{N}(0, \sigma^2_\Xi)$, when expressed in unit of length) is the Doppler measurement noise for the signal m from satellite s received by the user u.

Under the assumption that the multipath errors are negligible and absorbed by the measurement noise, the final form of the measurement equations, expressed in meters for pseudorange and carrier phase and m/s for Doppler, become

$$\begin{aligned}\underline{\Phi}^s_{u,m} = {} & \|\mathbf{r}_u - \mathbf{r}^s\| + dt^\Sigma_u - dt^s + m^s_{h,u} T_{\mathrm{h,u}} + m^s_{w,u} T_{\mathrm{w,u}} - \mu_m I^s_u + \lambda_m \beta_{u,m} - \lambda_m \beta^s_m \\ & + \zeta^s_{\mathrm{PCO,u,m}} + \zeta^s_{\mathrm{PCV,u,m}} + \lambda_m W^s_{\mathrm{u,m}} + \delta t^s_{\mathrm{Sha},u} + \Delta\rho^s_{u,m} + \lambda_m M^s_{u,m} + \underline{\varepsilon}^s_{u,m}\end{aligned} \tag{4}$$

$$\underline{P}^s_{u,m} = \|\mathbf{r}_u - \mathbf{r}^s\| + dt^\Sigma_u - dt^s + m^s_{h,u} T_{\mathrm{h},u} + m^s_{w,u} T_{\mathrm{w},u} + \mu_m I^s_u + b_{u,m} - b^s_m + \zeta^s_{\mathrm{PCO},u,m} + \delta t^s_{\mathrm{Sha},u} + \underline{\epsilon}^s_{u,m} \tag{5}$$

$$-\lambda_m \underline{d}^s_{u,m} = \underline{D}^s_{u,m} = \|\mathbf{v}_u - \mathbf{v}^s\| + \dot{dt}_u - \dot{dt}^s + \underline{\Xi}^s_{u.m} \tag{6}$$

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3.1.1. Ionosphere-free combination

In the framework of the computation of the SE PVT in the PP block, the ionosphere error is removed by using the ionosphere-free (IF) combination of the pseudorange measurement. The first signal adopted in the IF combination is the E1C for Galileo and the L1CA for GPS, while the second signal is one of the other selected signals, in order of preference E5bQ, E5aQ and E6B(C) for Galileo and L2C for GPS.

The rank-deficient Ionosphere-Free combination for the pseudorange measurements in unit of length is

$$\begin{aligned}\underline{P}^{s}_{u,\mathrm{if_j}} &= \frac{f_1^2}{f_1^2-f_j^2}\underline{P}^{s}_{u,1} - \frac{f_j^2}{f_1^2-f_j^2}\underline{P}^{s}_{u,j} \\ &= \|\mathbf{r}_u - \mathbf{r}^s\| + dt_u^{\Sigma} - dt^s + m^s_{h,u}T_{\mathrm{h},u} + m^s_{w,u}T_{\mathrm{w},u} + b_{u,\mathrm{if}_j} - b^s_{\mathrm{if}_j} + \zeta^s_{\mathrm{PCO},u,\mathrm{if}_j} + \delta t^s_{\mathrm{Sha},u} + \underline{\epsilon}^{s}_{u,\mathrm{if}_j}\end{aligned} \tag{7}$$

where the subscript if indicates the ionosphere-free combination, $\underline{(\cdot)}^{(s)}_{(u),\mathrm{if}_j} = \frac{f_1^2}{f_1^2-f_j^2}\underline{(\cdot)}^{(s)}_{(u),1} - \frac{f_j^2}{f_1^2-f_j^2}\underline{(\cdot)}^{(s)}_{(u),j}$, between the signal 1 (E1C for Galileo and L1CA for GPS) and the second signal j, for $j = 2, \dots, M_{\Sigma(s)}$.

3.2. HAS-UA PP

In the following subsections, a description of the HAS-UA PP steps is reported. The equations reported in this section and the rest of the document follow the notation defined in Appendix B.

3.2.1. Data Collection and Synchronization

Before executing the core algorithm, a preliminary step processes the input data including the selection of the relevant observations, retrieving the corresponding navigation data, and consolidating HAS corrections. Since corrections might arrive in separate blocks at different times, this step ensures that all applicable correction blocks are grouped together for each epoch, forming a complete correction set for the processed epoch.

3.2.2. Computation of satellite position, velocities, clock offset and clock drift from broadcast data

The selection of the BC data to be used is explained in Section 7 of the HAS SIS ICD [2] or Section 3.3 of the HAS IDD ICD [3]. In addition, the time of ephemeris of the BC data identified with the Reference IOD (IODnav for Galileo satellites and IODE/IODC for GPS satellites) shall have a time difference from the epoch of the GNSS measurements being processed that is not older than 21.3 hours for Galileo and 7 days for GPS.

The Galileo and GPS satellite positions, velocities, clock offsets and drifts are determined by the BC data as described by the respective Galileo OS SIS ICD [1] and GPS IS ICD [5]. The processing of satellites in view is performed as described by Figure 6 of the HAS SDD [4]: satellites in a healthy or marginal condition are treated if the corresponding valid HAS corrections are available.

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3.2.3. Computation of HAS corrected satellite positions, clock offsets and measurements

The HAS Initial Service corrections consist of orbit and clock corrections to the system BC data, and code and phase biases. These corrections can be applied as delta offset to be added to the satellite positions, velocities and clock errors obtained from the BC navigation data, as described in Section 3.2.2, and directly to the GNSS measurements, namely pseudorange and carrier phase measurements, in the case of the biases.

The HAS-UA applies the HAS corrections only to the pseudorange and carrier phase measurements while for the Doppler measurements only BC orbital and clock parameters are taken into account.

Before applying the HAS corrections, the HAS status, defined in [2] and [3], is also checked to verify that HAS is set to "operational mode" and that, consequently, nominal performance should be expected. On the contrary, if the HAS status is set to "do not use" the HAS corrections from all satellites should not be used and the previously received messages should be discarded. Moreover, satellites with data unavailability during the correction validity period are also discarded according to [2] and [3]. The reference time and reference frame used for Galileo HAS are the Galileo System Time (GST) and the Galileo Terrestrial Reference Frame (GTRF), respectively. The HAS corrections shall be applied as explained in Section 7 of the HAS SIS ICD [2] or Section 3.3 of the HAS IDD ICD [3].

The orbit during satellite shadow and eclipse periods may be considerably degraded and the removal of satellites under such conditions can improve the high-precision positioning results.

The eclipse condition can be defined considering a simple cylinder model for Earth's shadow as

$$\cos\phi = \frac{\tilde{\mathbf{r}}^s \cdot \mathbf{r}_{\text{Sun}}}{\|\tilde{\mathbf{r}}^s\|\|\mathbf{r}_{\text{Sun}}\|} < 0 \tag{8}$$

$$\|\tilde{\mathbf{r}}^s\|\sqrt{1-\cos^2\phi} < a_E \tag{9}$$

where $\mathbf{r}_{\text{Sun}}$ is the geocentric vector to the Sun, and a_E is the mean equatorial radius of Earth (6378137.0 m). ϕ is the angle between the geocentric vector to the satellite $\tilde{\mathbf{r}}^s$ and the geocentric vector to the Sun $\mathbf{r}_{\text{Sun}}$.

3.2.4. Computation of signal time of flight and satellite parameters at emission time

The measurements are linked to the signal reception time, which is given by the receiver time tags (i.e. in the receiver time scale), therefore an algorithm is needed to obtain the signal emission time in the HAS time scale. The GTRF frame is taken at reception time because it is a common reference for all measurements. In this section, two algorithms are presented, whose computation needs to be updated at every WLSE iteration as the receiver position and clock error estimation converges.

The first algorithm, which is iterative (hence the subscript i in $\tau_{u,i}^s$) and assumes that an approximate user position is known, is based on the following steps.

1. Set an approximate initial signal time travel $\tau_{u,1}^s$ (e.g. its value from previous epoch or 0.085/0.0885 s (Galileo/GPS) if $\mathbf{r}_u(t) = [0,0,0]^T$ or 0.0715/0.075 s if $\mathbf{r}_u(t)$ on the Earth surface).
2. Determine the BC satellite position $\mathbf{r}_{\text{BC}}^s(t)$, velocity $\mathbf{v}_{\text{BC}}^s(t)$, acceleration $\mathbf{a}_{\text{BC}}^s(t)$, clock error $dt_{BC}^s(t)$ (without relativistic correction) and clock drift $\dot{dt_{BC}^s}^s(t)$ at the time of reception $t = t' - \frac{dt_u^{\Sigma}(t)}{c}$, in GST and GPST time for Galileo and GPS respectively, from the broadcast ephemeris as indicated in Section 3.2.2.
3. Apply the HAS corrections to the satellite position and clock error computed in item 1 as described in Section 3.2.2 to obtain the HAS corrected satellite parameters: position $\mathbf{r}_{\text{APC}}^s(t)$, velocity $\mathbf{v}^s(t)$, acceleration $\mathbf{a}^s(t)$, clock error $dt^s(t)$ and clock drift $\dot{dt}^s(t)$ at the time of reception t.

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4. Apply the satellite s PCO correction to obtain the satellite s COM position $\mathbf{r}^s(t)$ as

$$\mathbf{r}^s(t) = \mathbf{r}^s_{\mathrm{APC}}(t) - \mathbf{A}^s \delta\mathbf{r}^s_{\mathrm{ORO},m}(t), \tag{10}$$

where $\mathbf{A}^s$ is the rotation matrix from the antenna reference frame to GTRF provided in Appendix C and $\delta\mathbf{r}^s_{\mathrm{ORO},m}(t)$ is the offset of the reference orbit (ORO), i.e. the PCO correction from the APC to which the HAS orbit is referred to the satellite COM.

5. Rotate $\mathbf{r}^s(t)$, $\mathbf{v}^s(t)$ and $\mathbf{a}^s(t)$ to the ECI coordinate system defined as being coincident with the GTRF coordinate system at the time $t + dt_u^\Sigma$, neglecting effects due to polar motion, nutation, and precession, obtaining

$$\mathbf{r}^s_{\mathrm{ECI}}(t) = \mathbf{R}_3(\dot\theta_E dt_u^\Sigma)\mathbf{r}^s(t) \tag{11}$$

$$\mathbf{v}^s_{\mathrm{ECI}}(t) = \dot{\mathbf{R}}_3\big(\dot\theta_E dt_u^\Sigma\big)\mathbf{r}^s(t) + \mathbf{R}_3(\dot\theta_E dt_u^\Sigma)\mathbf{v}^s(t) \tag{12}$$

$$\mathbf{a}^s_{\mathrm{ECI}}(t) = \ddot{\mathbf{R}}_3\big(\dot\theta_E dt_u^\Sigma\big)\mathbf{r}^s(t) + 2\dot{\mathbf{R}}_3\big(\dot\theta_E dt_u^\Sigma\big)\mathbf{v}^s(t) + \mathbf{R}_3(\dot\theta_E dt_u^\Sigma)\mathbf{a}^s(t) \tag{13}$$

where $\mathbf{R}_3(\theta)$, $\dot{\mathbf{R}}_3(\theta)$ and $\ddot{\mathbf{R}}_3(\theta)$ are the rotation matrices indicated in Appendix B and $\dot\theta_E$ is the Earth rotation rate indicated in the Galileo OS ICD.

6. Compute the geometrical range $\rho_u^s = \left\|\mathbf{r}^s_{\mathrm{ECI}}(t) - \mathbf{r}_{u,\mathrm{ECI}}(t)\right\|$.
7. Save the current signal time travel as $\tau^s_{u,i-1}$ and update the value of the travel time as $\tau^s_{u,i} = \frac{\rho_u^s}{c}$.
8. Compute the difference between the stored and the updated signal flight times, $\Delta\tau^s_{u,i} = \tau^s_{u,i} - \tau^s_{u,i-1}$.
9. Compute the satellite position at transmission time as $\mathbf{r}^s_{\mathrm{ECI}}(t-\tau^s_u) = \mathbf{r}^s_{\mathrm{ECI}}(t) - \tau^s_u\mathbf{v}^s_{\mathrm{ECI}}(t) - \frac{{\tau^s_u}^2}{2}\mathbf{a}^s_{\mathrm{ECI}}(t)$.
10. If the travel time divergence is higher than a determined threshold, $\Delta\tau^s_{u,i} > \epsilon_\tau$ (see Appendix F), and the iteration i is lower than the maximum allowed number of iterations $i_{\tau_{\max}}$, repeat items 6 to 9.
11. Compute the satellite velocity at transmission time as $\mathbf{v}^s_{\mathrm{ECI}}(t-\tau^s_u) = \mathbf{v}^s_{\mathrm{ECI}}(t) - \tau^s_u\mathbf{a}^s_{\mathrm{ECI}}(t)$.
12. Rotate the satellite velocity at transmission time back to the GTRF as

$$\tilde{\mathbf{r}}^s = \tilde{\mathbf{r}}^s(t-\tau^s_u) = \mathbf{R}_3^T(0)\mathbf{r}^s_{\mathrm{ECI}}(t-\tau^s_u) = \mathbf{r}^s_{\mathrm{ECI}}(t-\tau^s_u) \tag{14}$$

$$\tilde{\mathbf{v}}^s = \tilde{\mathbf{v}}^s(t-\tau^s_u) = \dot{\mathbf{R}}_3^T(0)\mathbf{r}^s_{\mathrm{ECI}}(t-\tau^s_u) + \mathbf{R}_3^T(0)\mathbf{v}^s_{\mathrm{ECI}}(t-\tau^s_u) \tag{15}$$

13. Compute the HAS satellite clock error $\widetilde{dt}^s(t-\tau^s_u)$ and drift $\widetilde{\dot{dt}}^s(t-\tau^s_u)$ at the transmission time as

$$\widetilde{dt}^s(t-\tau^s_u) = c\left(dt^s(t) - \tau^s_u\dot{dt}^s(t)\right) - \frac{2}{c}\left(\tilde{\mathbf{r}}^{sT}\tilde{\mathbf{v}}^s\right) \tag{16}$$

$$\widetilde{\dot{dt}}^s(t-\tau^s_u) = c\dot{dt}^s(t) \tag{17}$$

An alternative algorithm exploits the pseudorange measurements, using which the transmission time can be computed as

$$\tau^s_u = \frac{P^s_{u,1}(t)}{c} + dt^s(t) \tag{18}$$

(see Section 5.1.1.1 of [8]), and the satellite position at transmission time as

$$\tilde{\mathbf{r}}^s = \tilde{\mathbf{r}}^s(t-\tau^s_u) = \mathbf{R}_3\big(-\dot\theta_{\mathrm{E}}\,\tau^s_u\big)\mathbf{r}^s_{BC}(t-\tau^s_u) + \mathrm{R}^{\mathrm{ECEF}}_{\mathrm{NTW}}\delta\mathbf{r}^s_{\mathrm{HAS},m}(t-\tau^s_u) - \delta\mathrm{r}^s_{\mathrm{ORO},m}(t-\tau^s_u) \tag{19}$$

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3.2.5. Determination of receiver parameters with iterative WLSE

In the pre-processing phase, the SE PVT solution is estimated using only pseudorange and Doppler measurements and applying an iterative WLSE [17]. This solution is used for the initialization of the NAV solution, such as at the first epoch. Moreover, the timing solution from the SE PVT is also provided as output together with the solution from the NAV estimation block.

The objective is to estimate the user coordinates and the Galileo and GPS clock offsets by applying the SE PVT technique to the pseudorange measurements of at least four or five satellites for respectively the single- and dual-constellation cases. For this purpose, first a screening of the measurements to be processed is performed. In particular, as baseline, only measurements from satellites with elevations above tr_{co} cut-off threshold and with a signal C/N0 above tr_{cn0} threshold are processed (their values are provided in Annex F). As explained in Section 3.2.6, a post-estimation outlier detection check is also performed.

3.2.5.1. SE PVT processing

The iterative WLSE technique (the subscript i indicates the WLSE iteration) estimates the updates of the state vector $\delta\bar{\boldsymbol{x}}_{u,i}$ by minimizing the sum of the square residuals as indicated by the following expression

$$\delta\hat{\underline{\mathbf{x}}}_{u,i} = \arg\min_{\delta\bar{\mathbf{x}}_u \in \mathbb{R}^n} \left\| \delta\underline{\tilde{\mathbf{y}}}_{u,i} - \mathbf{H}_i \delta\bar{\mathbf{x}}_{u,i} \right\|^2_{\boldsymbol{\Sigma}^{-1}_{\tilde{y}\tilde{y}}} \tag{20}$$

with corresponding squared norm $\|\mathbf{a}\|^2_W = \mathbf{a}^T\mathbf{W}\mathbf{a}$. Then, the estimated solution $\delta\hat{\underline{\mathbf{x}}}_{u,i}$ can be obtained by

$$\delta\hat{\underline{\mathbf{x}}}_{u,i} = \left(\mathbf{H}_i{}^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}_{\tilde{y}\tilde{y}}\mathbf{H}_i\right)^{-1}\mathbf{H}_i^T\boldsymbol{\Sigma}^{-1}_{\tilde{y}\tilde{y}}\delta\underline{\tilde{\mathbf{y}}}_{u,i} \tag{21}$$

with

$$\boldsymbol{\Sigma}_{\delta\hat{\mathbf{x}}_i} = \left(\mathbf{H}_i^T\boldsymbol{\Sigma}^{-1}_{\tilde{y}\tilde{y}}\mathbf{H}_i\right)^{-1} \tag{22}$$

as its covariance matrix.

The user state vector can be computed by updating the previously available estimated vector $\boldsymbol{x}_{u,i-1}$ with the estimated variations $\boldsymbol{\delta}\hat{\mathbf{x}}_{u,i}$ as in the following

$$\hat{\underline{\mathbf{x}}}_{\mathrm{u},i} = \hat{\underline{\mathbf{x}}}_{u,i-1} + \delta\hat{\underline{\mathbf{x}}}_{u,i} \tag{23}$$

Therefore, the observations are linearized around this updated vector and the solution can be iterated until convergence. The pseudorange and Doppler expressions are non-linear with respect to the position and velocity parameters in the state vector: a linearization of the measurement around an approximate solution is performed in Equations (24) and (25) and the solution is solved iteratively [6]. The linearization is performed using a Taylor expansion truncated at the first order, as detailed in [6].

The observed minus computed pseudorange $\delta\underline{\tilde{P}}^s_{u,\mathrm{if},i}$ and Doppler $\delta\underline{\tilde{D}}^s_{u,m,i}$ measurements used in the SE PVT estimation follow from Equations (5) and (6) as

$$\delta\underline{\tilde{P}}^s_{u,\mathrm{if},i} = \underline{P}^s_{u,\mathrm{if}} - \tilde{P}^s_{u,\mathrm{if},i-1} = \left(\frac{\partial\tilde{\mathrm{P}}^s_{u,\mathrm{if}}}{\partial\bar{\mathbf{x}}_u}\bigg|_{\bar{\mathbf{x}}_u=\bar{\mathbf{x}}_{u,i-1}}\right)^T\delta\bar{\mathbf{x}}_u \tag{24}$$

$$\delta\underline{\tilde{D}}^s_{u,m,i} = \underline{D}^s_{u,m} - \tilde{D}^s_{u,m,i-1} = \left(\frac{\partial\tilde{D}^s_{u,m}}{\partial\bar{\mathbf{x}}_u}\bigg|_{\bar{\mathbf{x}}_u=\bar{\mathbf{x}}_{u,i-1}}\right)^T\delta\bar{\mathbf{x}}_u \tag{25}$$

where $\tilde{P}^s_{u,\mathrm{if,i}}$ and $\tilde{\mathrm{D}}^s_{u,m,i}$ are the computed measurements, the subscript i indicates the iteration of the computation, the superscript $\cdot^T$ indicates the transpose operation and $\bar{\boldsymbol{x}}_u$ is the SE PVT user state vector,

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estimated through WLSE, while $\delta\overline{\boldsymbol{x}}_u$ is its variation. The subscripts i and $i-1$ indicate that the parameter is referred to the WLSE iteration i or $i-1$ respectively. The SE PVT user state vector $\overline{\boldsymbol{x}}_u$, is defined as

$$\bar{\mathbf{x}}_u = \left[\bar{\mathbf{r}}_u^T, \bar{\mathbf{v}}_u^T, \overline{\mathbf{dt}}_u^T, \overline{\dot{dt}}_u\right]^T \tag{26}$$

where

- $\bar{\mathbf{r}}_u^T = \mathbf{r}_u$ is the vector of the estimable user receiver coordinates in the primary GNSS coordinate system,
- $\bar{\mathbf{v}}_u^T = \mathbf{v}_u$ is the vector of the estimable user receiver velocity in the primary GNSS coordinate system,
- $\overline{\mathbf{dt}}_u^T = \left[\overline{dt}_u^E,\ \overline{dt}_u^G\right]^T$ is the vector of the estimable receiver clock offsets w.r.t. the Galileo and GPS system times (estimable indicates that the clock also contains the lumped receiver biases of the reference frequencies and the system network parameters absorbed from the applied corrections),
- $\overline{\dot{dt}}_u = \dot{dt}_u$ is the estimable user receiver clock drift,

while $\delta\bar{\mathbf{x}}_u$ is the variation of the user state vector defined as

$$\delta\bar{\mathbf{x}}_u = \left[\delta\bar{\mathbf{r}}_u^T, \delta\bar{\mathbf{v}}_u^T, \delta\overline{\mathbf{dt}}_u^T, \delta\overline{\dot{dt}}_u\right]^T \tag{27}$$

The computed measurements can be expressed as

$$\tilde{P}^s_{u,\mathrm{if}_j,\mathrm{i}-1} = \mathbf{e}_u^s\left(\bar{\mathbf{r}}_{u,i-1} - \tilde{\mathbf{r}}^s\right) + \overline{dt}^{\Sigma}_{u,i-1} - \widetilde{dt}^s + \tilde{m}^s_{\mathrm{h},u}\tilde{T}_{\mathrm{h},u} + \tilde{m}^s_{\mathrm{w},u}\tilde{T}_{\mathrm{w},u} - \tilde{b}^s_{\mathrm{if}_j} + \tilde{\zeta}^s_{\mathrm{PCO},u,\mathrm{if}} \tag{28}$$

$$\widetilde{D}^s_{u,m,i-1} = \mathbf{e}_u^s\left(\bar{\mathbf{v}}_{u,i-1} - \tilde{\mathbf{v}}^s\right) + \overline{\dot{dt}}_{u,i-1} - \widetilde{\dot{dt}}^s + \underline{\Xi}^s_{u.m} \tag{29}$$

where

- $\widetilde{\mathbf{r}^s}$ indicates the satellite position obtained by the HAS orbits, as described in Section 3.2.4;
- $\widetilde{dt}^s$ is the HAS clock error for satellite s, including the relativistic correction;
- $\tilde{b}^s_m$ is the HAS satellite code bias for satellite s and signal m;
- $\tilde{m}^s_{\mathrm{w},u}$ and $\tilde{m}^s_{\mathrm{h},u}$ are respectively the wet and hydrostatic Neill tropospheric mapping functions [11];
- $\tilde{T}_{\mathrm{w},u}$ and $\tilde{T}_{\mathrm{h},u}$ are respectively the wet and hydrostatic zenith tropospheric corrections (SBAS RTCA-MOPS) [14];
- $\widetilde{\dot{dt}}^s$ is the clock drift for satellite s, including the relativistic correction.

The resulting estimation system, referred to a generic temporal instant t (whose index is omitted for simplicity) is obtained substituting Eq. (30) and (31) in Eq. (24) and (25), leading to

$$\begin{aligned}&\delta\underline{\tilde{P}}^s_{u,\mathrm{if},i} = \frac{f_1^2}{f_1^2 - f_m^2}\underline{P}^s_{u,1} - \frac{f_m^2}{f_1^2 - f_m^2}\underline{P}^s_{u,m} - \mathbf{e}_u^s\left(\hat{\mathbf{r}}_{u,i-1} - \tilde{\mathbf{r}}^s\right) - \widehat{dt}^{\Sigma}_{u,i-1} + \widetilde{dt}^s + \tilde{b}^s_m \\ &-\tilde{m}^s_{\mathrm{h},u}\tilde{T}_{\mathrm{h},u} - \tilde{m}^s_{\mathrm{w},u}\tilde{T}_{\mathrm{w},u} - \tilde{\zeta}^s_{\mathrm{PCO},u,m} = \mathbf{e}_u^s\delta\bar{\mathbf{r}}_{u,i} + \left(1 - \frac{\|\mathbf{v}_u^s\|}{c}\right)\delta\overline{dt}^{\Sigma}_{u,i} + \underline{\epsilon}^s_{u,m}\end{aligned} \tag{30}$$

$$\delta\underline{\widetilde{D}}^s_{u,m,i} = \underline{D}^s_{u,m} - \mathbf{e}_u^s\left(\hat{\mathbf{v}}_{u,i-1} - \tilde{\mathbf{v}}^s\right) - \widehat{\dot{dt}}_{u,i-1} + \widetilde{\dot{dt}}^s = \frac{\mathbf{v}_u^s - (\mathbf{v}_u^s\mathbf{e}_u^s)\mathbf{e}_u^s}{\|\hat{\mathbf{r}}_{u,i-1} - \tilde{\mathbf{r}}^s\|}\delta\bar{\mathbf{r}}_{u,i} + \mathbf{e}_u^s\delta\bar{\mathbf{v}}_{\mathrm{u,i}} + \delta\overline{\dot{dt}}_{u,i} + \underline{\Xi}^s_{u.m} \tag{31}$$

where

- the subscript $i-1$ indicates that the value of the estimable parameter is referred to the previous WLSE iteration or epoch solution or at its initialisation;
- the subscript i indicates that the value of the estimable parameter variation is referred to the current iteration and epoch;

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- $\delta\overline{\boldsymbol{r}}_{u,i}$ indicates the variation of the estimable user receiver position;
- $\delta\overline{dt}_{u,i}^{\Sigma}$ indicates the variation of the estimable user receiver clock error for system Σ;
- $\delta\overline{\boldsymbol{v}}_{\mathrm{u,i}}$ indicates the variation of the estimable user receiver velocity;
- $\delta\overline{\dot{dt}}_{u,i}$ indicates the variation of the estimable derivative of the user receiver clock error.

The linear model relating $\delta\bar{\mathbf{x}}_{u,m}$ to the pseudorange and Doppler measurements can be expressed in matrix form as

$$\delta\underline{\tilde{\mathbf{y}}}_{u,i} = \mathbf{H}_i\delta\bar{\mathbf{x}}_{u,i} + \underline{\boldsymbol{\varepsilon}}_u \tag{32}$$

where

- $\delta\underline{\tilde{\mathbf{y}}}_{u,\mathrm{i}} = \left[\delta\underline{\tilde{\mathbf{p}}}_{u,i}^{E\ T}, \delta\underline{\tilde{\mathbf{p}}}_{u,i}^{G\ T}, \delta\underline{\tilde{\mathbf{d}}}_{u,i}^{E\ T}, \delta\underline{\tilde{\mathbf{d}}}_{u,i}^{G\ T}\right]^T$ is the vector of the differences between the measured and computed measurements,
- $\delta\underline{\tilde{\mathbf{p}}}_{u,i}^{\Sigma} = \left[\delta\underline{\tilde{P}}_{u,1,i}^{1}, \dots, \delta\underline{\tilde{P}}_{u,1,i}^{S_\Sigma}, \delta\underline{\tilde{P}}_{u,2,i}^{1}, \dots, \delta\underline{\tilde{P}}_{u,2,i}^{S_\Sigma} \dots, \delta\underline{\tilde{P}}_{u,\mathrm{M}_\Sigma,i}^{1}, \dots, \delta\underline{\tilde{P}}_{u,\mathrm{M}_\Sigma,i}^{S_\Sigma}\right]^T$ is the vector of the differences between the measured and computed pseudoranges for system Σ, with $\Sigma = E \mid G$,
- $\delta\underline{\tilde{\mathbf{d}}}_{u,i}^{\Sigma} = \left[\delta\underline{\tilde{D}}_{u,1,i}^{1}, \dots, \delta\underline{\tilde{D}}_{u,1,i}^{S_\Sigma}, \delta\underline{\tilde{D}}_{u,2,i}^{1}, \dots, \delta\underline{\tilde{D}}_{u,2,i}^{S_\Sigma} \dots, \delta\underline{\tilde{D}}_{u,\mathrm{M}_\Sigma,i}^{1}, \dots, \delta\underline{\tilde{D}}_{u,\mathrm{M}_\Sigma,i}^{S_\Sigma}\right]^T$ is the vector of the differences between the measured and computed Doppler for system Σ, with $\Sigma = E \mid G$,
- M_Σ is the number of signals of the system $\Sigma = E|G$ used by the user,
- $M = M_E + M_G$ is the total number of signals used by the user,
- S_Σ is the number of satellites of the system $\Sigma = E|G$ used by the user,
- $S = S_E + S_G$ is the total number of satellites used by the user,
- $\mathbf{H}_i$ is the design (or geometry) matrix used to project the measurements into the state vector space,
- $\underline{\boldsymbol{\varepsilon}}_u$ is the vector of the measurement noises.

The design matrix $\mathbf{H}_i$ can be expressed as follows

$$\mathbf{H}_i = \begin{bmatrix} \mathbf{H}_{P,r}^{E} & \mathbf{0} & \mathbf{H}_{P,dt^E}^{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{H}_{P,r}^{G} & \mathbf{0} & \mathbf{0} & \mathbf{H}_{P,dt^G}^{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{D,v}^{E} & \mathbf{0} & \mathbf{0} & \mathbf{H}_{D,\dot{dt}}^{E} \\ \mathbf{0} & \mathbf{H}_{D,v}^{G} & \mathbf{0} & \mathbf{0} & \mathbf{H}_{D,\dot{dt}}^{G} \end{bmatrix} \tag{33}$$

$$\mathbf{H}_{P,r}^{\Sigma} = \mathbf{H}_{D,v}^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} \mathbf{e}_u^{1\ T} \\ \vdots \\ \mathbf{e}_u^{S_\Sigma T} \end{bmatrix}, \text{ for } \Sigma = E|G, \tag{34}$$

$$\mathbf{H}_{P,dt^\Sigma}^{\Sigma} = \mathbf{H}_{D,\dot{dt}}^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \mathbf{1}_{S_\Sigma}, \text{ for } \Sigma = E|G, \tag{35}$$

where

- $\mathbf{H}_{O,p}^{\Sigma}$ contains the partial derivatives of the observable $O = \{P, D\}$ from GNSS $\Sigma = \{E, G\}$ w.r.t. the parameter p,$= \{r, v, dt^\Sigma, \dot{dt}\}$
- $\mathbf{1}_a = \underbrace{[1 \dots 1]}_{a \text{ times}}^T$,
- $\mathbf{C}_{a,b} = \begin{bmatrix} \mathbf{0}_{a\times b-a} \\ \mathbf{I}_{b-a} \end{bmatrix}$,
- $\mathbf{I}_a = \mathrm{diag}\underbrace{(1 \dots 1)}_{a \text{ times}}$,

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- $\otimes$ is the Kronecker product.

The vector of observed minus computed measurements $\delta\tilde{\mathbf{y}}_{u,i} \sim N\left(\mathbf{H}\delta\bar{\mathbf{x}}_{u,i}, \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}\right)$ can be expressed in matrix form as

$$\delta\underline{\tilde{\mathbf{y}}}_{u,i} = \mathbf{T}\left[\underline{\hat{\mathbf{p}}}_{u,i}^{T}, \underline{\hat{\mathbf{d}}}_{u,i}^{T}, \tilde{\mathbf{r}}^{S^T}, \tilde{\mathbf{v}}^{S^T}, \widetilde{\mathbf{dt}}^{S^T}, \widetilde{\dot{\mathbf{dt}}}^{S^T}, \tilde{\mathbf{b}}_{M}^{S^T}, \tilde{T}_{H,u}, \tilde{T}_{W,u}, \tilde{\boldsymbol{\zeta}}_{PCO,u}^{S}\right]^{\boldsymbol{T}} \tag{36}$$

$$\mathbf{T} = \begin{bmatrix} \mathbf{F}^{\mathbf{S}} & \mathbf{0} & \mathbf{U}_u^S & \mathbf{0} & \mathbf{I}_n & \mathbf{0} & \mathbf{I}_n & -\mathbf{m}_{H,u} & -\mathbf{m}_{W,u} & -\mathbf{I}_n \\ \mathbf{0} & \mathbf{I}_n & \mathbf{0} & \mathbf{U}_u^S & \mathbf{0} & \mathbf{I}_n & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{37}$$

with $n = M_E S_E + M_G S_G$ and where (although equations are provided for generic M_E and M_G values, in SE PVT they are set as $M_E = M_G = 2$)

- $\underline{\hat{\mathbf{p}}}_{u,i} = \begin{bmatrix} \underline{\mathbf{p}}_u^E \\ \underline{\mathbf{p}}_u^G \end{bmatrix} - \begin{bmatrix} \mathbf{H}_{P,r}^E \\ \mathbf{H}_{P,r}^G \end{bmatrix} \hat{\mathbf{r}}_{u,i-1} - \begin{bmatrix} -\widehat{dt}_{u,i-1}^E \mathbf{1}_{M_E S_E} \\ -\widehat{dt}_{u,i-1}^G \mathbf{1}_{M_G S_G} \end{bmatrix}$,
- $\underline{\mathbf{p}}_{u,i}^{\Sigma} = \left[\underline{P}_{u,1,i}^1, \dots, \underline{P}_{u,1,i}^{S_\Sigma}, \underline{P}_{u,2,i}^1, \dots, \underline{P}_{u,2,i}^{S_\Sigma} \dots, \underline{P}_{u,\mathrm{M}_\Sigma,i}^1, \dots, \underline{P}_{u,\mathrm{M}_\Sigma,i}^{S_\Sigma}\right]^T$ is the vector of the pseudorange measurements for system Σ, with $\Sigma = E \mid G$,
- $\underline{\hat{\mathbf{d}}}_{u,i} = \begin{bmatrix} \underline{\mathbf{d}}_u^E \\ \underline{\mathbf{d}}_u^G \end{bmatrix} - \begin{bmatrix} \mathbf{H}_{D,r}^E \\ \mathbf{H}_{D,r}^G \end{bmatrix} \hat{\mathbf{r}}_{u,i-1} - \begin{bmatrix} \mathbf{H}_{D,v}^E \\ \mathbf{H}_{D,v}^G \end{bmatrix} \hat{\mathbf{v}}_{u,i-1} - \hat{d}_{u,i-1}\mathbf{1}_n$,
- $\underline{\mathbf{d}}_{u,i}^{\Sigma} = \left[D_{u,1,i}^1, \dots, \underline{D}_{u,1,i}^{S_\Sigma}, D_{u,2,i}^1, \dots, D_{u,2,i}^{S_\Sigma} \dots, \underline{D}_{u,\mathrm{M}_\Sigma,i}^1, \dots, \underline{D}_{u,\mathrm{M}_\Sigma,i}^{S_\Sigma}\right]^T$ is the vector of the Doppler measurements for system Σ, with $\Sigma = E \mid G$,
- $\tilde{\mathbf{r}}^S = \left[\tilde{\mathbf{r}}^{E^T}, \tilde{\mathbf{r}}^{G^T}\right]^T$, $\tilde{\mathbf{r}}^{\Sigma} = \left[\tilde{\mathbf{r}}^{1^T}, \dots, \tilde{\mathbf{r}}^{S_\Sigma{}^T}\right]^T$,
- $\tilde{\mathbf{v}}^S = \left[\tilde{\mathbf{v}}^{E^T}, \tilde{\mathbf{v}}^{G^T}\right]^T$, $\tilde{\mathbf{v}}^{\Sigma} = \left[\tilde{\mathbf{v}}^{1^T}, \dots, \tilde{\mathbf{v}}^{S_\Sigma{}^T}\right]^T$,
- $\widetilde{\mathbf{dt}}^S = \left[\widetilde{\mathbf{dt}}^{E^T}, \widetilde{\mathbf{dt}}^{G^T}\right]^T$, $\widetilde{\mathbf{dt}}^{\Sigma} = \left[\widetilde{dt}^1, \dots, \widetilde{dt}^{S_\Sigma}\right]^T$,
- $\widetilde{\dot{\mathbf{dt}}}^S = \left[\widetilde{\dot{\mathbf{dt}}}^{E^T}, \widetilde{\dot{\mathbf{dt}}}^{G^T}\right]^T$, $\widetilde{\dot{\mathbf{dt}}}^{\Sigma} = \left[\widetilde{\dot{dt}}^1, \dots, \widetilde{\dot{dt}}^{S_\Sigma}\right]^T$,
- $\tilde{\mathbf{b}}_M^S = \left[\tilde{\mathbf{b}}^{E^T}, \tilde{\mathbf{b}}^{G^T}\right]^T$, $\tilde{\mathbf{b}}^{\Sigma} = \left[\tilde{b}_1^1, \dots, \tilde{b}_1^{S_\Sigma}, \dots, \tilde{b}_{M_\Sigma}^1, \dots, \tilde{b}_{M_\Sigma}^{S_\Sigma}\right]^T$,
- $\tilde{\boldsymbol{\zeta}}_{PCO,u}^S = \left[\tilde{\boldsymbol{\zeta}}_{PCO,u}^{E\,T}, \tilde{\boldsymbol{\zeta}}_{PCO,u}^{G\,T}\right]^T$, $\tilde{\boldsymbol{\zeta}}_{PCO}^{\Sigma} = \left[\tilde{\zeta}_{PCO,u,1}^1, \dots, \tilde{\zeta}_{PCO,u,1}^{\mathrm{S}_\Sigma}, \dots, \tilde{\zeta}_{PCO,u,M_\Sigma}^1, \dots, \tilde{\zeta}_{PCO,u,M_\Sigma}^{\mathrm{S}_\Sigma}\right]^T$
- $\mathbf{F}^{\mathrm{S}} = \begin{bmatrix} \mathbf{F}^E & \mathbf{0} \\ \mathbf{0} & \mathbf{F}^G \end{bmatrix}$, $\mathbf{F}^{\Sigma} = \begin{bmatrix} \frac{f_{1_\Sigma}^2}{f_{1_\Sigma}^2 - f_{2_\Sigma}^2} \boldsymbol{I}_{S_\Sigma} & \frac{-f_{2_\Sigma}^2}{f_{1_\Sigma}^2 - f_{2_\Sigma}^2} \boldsymbol{I}_{S_\Sigma} & \dots & \frac{-f_{M_\Sigma}^2}{f_{1_\Sigma}^2 - f_{M_\Sigma}^2} \boldsymbol{I}_{S_\Sigma} \end{bmatrix}$
- $\mathbf{U}_u^S = \begin{bmatrix} \mathbf{U}_u^E & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_u^G \end{bmatrix}$, $\mathbf{U}_u^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \left[\bigoplus_{s=1}^{S_\Sigma} \mathbf{e}_u^{s\,T}\right]$
- $\mathbf{m}_{\ni,u}^S = \begin{bmatrix} \mathbf{m}_{\ni,\mathrm{u}}^E \\ \mathbf{m}_{\ni,\mathrm{u}}^G \end{bmatrix}$, for $\ni = \mathrm{H} \mid \mathrm{W}$, $\mathbf{m}_{\ni,u}^{\Sigma} = \left[m_{\ni,\mathrm{u}}^1, \dots, m_{\ni,\mathrm{u}}^{S_\Sigma}\right]^T$.

Its first two moments are

$$E\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i}\right) = \mathbf{H}_i \delta\bar{\mathbf{x}}_{u,i} \tag{38}$$

$$D\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i}\right) = \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}} = \mathbf{T}\begin{bmatrix} \boldsymbol{\Sigma}_{\overline{yy}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{cc} \end{bmatrix}\mathbf{T}^T \tag{39}$$

where $E\left(\underline{y}\right)$ is the expectation of the random variable $\underline{y}$ and $D\left(\underline{y}\right)$ is its dispersion,

- $\boldsymbol{\Sigma}_{\overline{yy}} = \mathrm{diag}\left(\boldsymbol{\Sigma}_{\overline{PP}}^E, \boldsymbol{\Sigma}_{\overline{PP}}^G, \boldsymbol{\Sigma}_{\overline{DD}}^E \boldsymbol{\Sigma}_{\overline{DD}}^G\right)$.
- $\boldsymbol{\Sigma}_{\overline{PP}}^{\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} {\sigma_{P,u,m}^s}^2$ is the variance matrix of the pseudorange measurements,

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- $\boldsymbol{\Sigma}_{\overline{DD}}^{\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} {\sigma_{D,u,m}^{s}}^2$ is the variance matrix of the Doppler measurements,
- $\bigoplus$ is the direct sum operator (see Appendix B),
- $\sigma_{P,u,m}^{s} = m_{P,u}^{s}\sigma_{P,u,m}$ and $\sigma_{D,u,m}^{s} = m_{D,u}^{s}\sigma_{D,u,m}$ are the standard deviations of pseudorange and Doppler measurements for the signal m from the satellite s generated by the user receiver u,
- $m_{P,u}^{s} = m_{D,u}^{s} = 1/sin^2\,(\theta_u^s)$ are the mapping function for respectively the variance of the pseudorange and Doppler measurements,
- θ_u^s is the satellite s elevation,
- $\sigma_{P,u,m}$ and $\sigma_{D,u,m}$ are respectively the standard deviations of the pseudorange and Doppler measurements for the signal m from a hypothetical satellite at zenith,

If the full $\boldsymbol{\Sigma}_{cc}$ matrix is not available, it can be ignored for SE PVT.

Finally, the WLSE technique is applied [6], by exploiting that the WLSE of $\delta\bar{\mathbf{x}}_{u,i}$ is identical to the BLUE of $\delta\bar{\mathbf{x}}_{u,i}$ when the weight matrix is chosen equal to the inverse of the proper variance matrix of the vector of observables (not always true in practice), i.e.

$$\mathbf{W} = \boldsymbol{\Sigma}_{\bar{y}\bar{y}}^{-1} \tag{40}$$

The above matrix allows taking into consideration that the noise affecting the different measurements is not the same. The numerical values for the parameters listed in this section are provided in Appendix F.

3.2.5.2. SE PVT initialisation

At the very beginning of the positioning computation the user state vector and its variance are considered completely unknown. For this purpose, a simple iterative LS estimation is performed at the first iteration of the first epoch, in order to determine an initial approximated solution. At this stage, the user is positioned statically at the centre of the reference system, hence at the centre of the Earth, and no troposphere correction is applied on the pseudorange measurements or satellite removed because of cut-off thresholds (e.g. elevation or C/N0). The observed minus computed measurements for the initialisation are provided by Eq. (30) and (31) with initially $i = 1$, $\hat{\mathbf{x}}_{\mathbf{u,0}} = \mathbf{0}$. The estimated solution $\delta\hat{\boldsymbol{x}}_{u,i1}$ can be obtained by

$$\boldsymbol{\delta}\underline{\hat{\mathbf{x}}}_{u,1} = (\mathbf{H}_1^T\mathbf{H}_1)^{-1}\mathbf{H}_1^T\delta\underline{\tilde{\mathbf{y}}}_{u,1} \tag{41}$$

with

$$\boldsymbol{\Sigma}_{\delta\hat{\boldsymbol{x}}_1} = (\mathbf{H}_1^T\mathbf{H}_1)^{-1}\mathbf{H}_1^T\boldsymbol{\Sigma}_{\bar{y}\bar{y}}\mathbf{H}_1(\mathbf{H}_1^T\mathbf{H}_1)^{-1} \tag{42}$$

as its covariance matrix. The solution of the first iteration, $\underline{\hat{\boldsymbol{x}}}_{u,i1}$, $\boldsymbol{\Sigma}_{\hat{\boldsymbol{x}}_{\boldsymbol{i1}}}$, can be provided to the WLSE filter.

3.2.5.3. SE PVT stop criteria

The WLSE stopping criteria is based on the inner product of the user state vector increment

$$\left\|\boldsymbol{\delta}\underline{\hat{\mathbf{x}}}_{u,i}\right\|_{\boldsymbol{\Sigma}_{\delta\hat{\boldsymbol{x}}_{\boldsymbol{i}}}}^2 < \mathrm{tr}_{\boldsymbol{\delta}\underline{\hat{\mathbf{x}}}_{u,i}} \tag{43}$$

A second stop criteria shall be also used based on the number of performed iterations. This number should be decided by the user based on its computational power and anyhow should be kept within $\mathrm{tr}_{\mathrm{WLS}_i}$ iterations. As soon as the WLSE iterations reach this number the SE PVT estimation should be stopped (see Appendix F for the numerical values).

As soon as one of these criteria is met, the iterative WLSE shall be stopped.

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3.2.6. Outlier detection and adaptation

During the pre-processing phase, the HAS corrections and the measurements are checked for the identification and removal of outliers using the Detection, Identification and Adaption (DIA) procedure. DIA performs hypothesis testing of the linear model adopted for the SE PVT estimation determining if the observed process is better described by an alternative model. This test helps to identify and remove outliers by comparing different model hypotheses. As already specified before, the measurements are assumed to have a normal distribution and their stochastic model is identical in all the hypotheses considered, which are

$$H_o: E\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i}\right) = \mathbf{H}_i\delta\bar{\mathbf{x}}_{u,i} \tag{44}$$

and

$$H_a: E\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i}\right) = \mathbf{H}_i\delta\bar{\mathbf{x}}_{u,i} + \mathbf{C}_\mathrm{y}\boldsymbol{\nabla} \tag{45}$$

with $\boldsymbol{\nabla}$ a q-vector of additional unknown parameters in the alternative hypotheses related to the expectation of $\delta\underline{\hat{\mathbf{y}}}_{u,i}$ by the $h \times q$ matrix $\mathbf{C}_\mathrm{y}$ (also assumed full rank, $\mathrm{rank}(\mathbf{C}_\mathrm{y}) = \mathrm{q}, 1 \le q \le h - l$), where q is the number of additional unknown parameters in the alternative hypotheses (also called degree of freedom), h is the total number of measurements and l is the number of parameters in the state vector.
It is worth underlining that the proposed outlier detection method is based on a simple approach aimed to identify only a single failure at a time. More elaborated and robust strategies should be adopted according to the user-specific application or working environment. Therefore, the user should be aware of this aspect and, if necessary, exploit alternative methods to satisfy the target performance. Starting from the estimation of the SE PVT solution obtained as described in Section 3.2.4 and the evaluation of the least-square residual vector $\hat{\boldsymbol{e}}_{u,i}$ given by

$$\underline{\hat{\mathbf{e}}}_{u,i} = \delta\underline{\tilde{\mathbf{y}}}_{u,i} - \mathbf{H}_i\delta\underline{\hat{\mathbf{x}}}_{u,i} = \delta\underline{\tilde{\mathbf{y}}}_{u,i} - \delta\underline{\hat{\mathbf{y}}}_{u,i} \tag{46}$$

with variance

$$\boldsymbol{\Sigma}_{\hat{e}\hat{e}} = \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}} - \boldsymbol{\Sigma}_{\hat{y}\hat{y}} = \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}} - \mathbf{H}_i\boldsymbol{\Sigma}_{\delta\hat{x}}\mathbf{H}_i^T = \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}} - \mathbf{H}_i\left(\mathbf{H}_i^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{H}_i\right)^{-1}\mathbf{H}_i^T \tag{47}$$

the adopted method is based on the following steps.

1. ***Detection***. It is performed by adopting the estimator of the variance factor of unit weight, $\underline{\hat{\sigma}}^2$, where $q = h - l$, i.e. there is no redundancy in the alternative hypothesis H_a, implying that the least square residuals of the alternative hypothesis are $\underline{\hat{\mathbf{e}}}_{u,a,i} = 0$. The expression of the $\underline{\hat{\sigma}}^2$-test statistic is

$$\underline{\hat{\sigma}}^2 = \frac{\underline{T}_{q=h-l}}{h-l} = \frac{\left\|\underline{\hat{\mathbf{e}}}_{u,i}\right\|^2_{\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}}}{h-l} \sim F(h-l, \infty, \lambda) \tag{48}$$

with non-centrality parameter $\lambda = 0$ for H_0 and where $\underline{T}_{q=h-l} \sim \chi^2(h-l, \lambda)$ is the overall model test. The null hypothesis H_0, corresponding to the nominal model described in Section 3.2.5.1, is rejected if

$$\underline{\hat{\sigma}}^2 > k^F_{\alpha_\sigma} = \frac{k^{\chi^2}_{\alpha_\sigma}}{h-l} \tag{49}$$

where $k^F_{\alpha_\sigma}$ is the critical value for the F-distribution once that the probability of false alarm α_σ has been fixed. For numerical reasons it is more convenient to use the χ^2-distribution for the computation of the critical value using the relation in Eq. (49) (please note that this equality is valid only for the critical values, not for the distributions in general).

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2. ***Identification***. If the $\underline{\hat{o}}^2$ test rejects the nominal model H_0, then it is necessary to identify where the error lies. Now a wide range of alternative hypotheses is considered since, at each time, an outlier is assumed to be present in just one of the observations and this is done for each observation. All the alternative hypotheses extend the null hypothesis with just a single ($q = 1$) additional unknown parameter ∇, so that $\mathbf{C}_y = \mathbf{c}_j = \left[\underbrace{0, \dots, 0}_{j-1}, 1, \underbrace{0, \dots, 0}_{h-j}\right]^T$ and

$$H_{a_j}: E\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i}\right) = \mathbf{H}_i\delta\bar{\mathbf{x}}_{u,i} + \mathbf{c}_j\boldsymbol{\nabla}, \text{ for } j = 1, \dots, h \tag{50}$$

The h alternative hypotheses are tested using the w-test

$$\underline{w} = \frac{\mathbf{c}_j^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\underline{\hat{\mathbf{e}}}_{u,i}}{\sqrt{\mathbf{c}_j^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\boldsymbol{\Sigma}_{\hat{e}\hat{e}}^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{c}_j}} \tag{51}$$

which for H_0 has distribution $N(1,0)$. If $\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}$ is diagonal (but please note that in the generic case presented in Section 3.2.5.1 it is not) then the w-test checking for the *j-th* observation reduces to

$$\underline{w}_j = \frac{\hat{e}_{a_j,i}}{\sigma_{\hat{e}_{a_j,i}}} \tag{52}$$

(which is also known as normalised residuals) with $\hat{e}_{a_j,i} = \boldsymbol{c}_j^T\hat{\boldsymbol{e}}_{u,i}$. The maximum is taken over the h different hyphotesis $|w_a|$, for $a = 1, \dots, h$, with

$$\max_{j=1,\dots,h}|w_j| = |w_k| \tag{53}$$

verifying the condition

$$|w_k| > \sqrt{k_{\frac{\alpha_w}{2}}^N}, \text{ where } \sqrt{k_{\frac{\alpha_w}{2}}^N} = \sqrt{k_{\alpha_w}^{\chi^2}} \tag{54}$$

The error model is assigned to the observation having the largest outlier test statistic exceeding the critical value. The probability of deciding for H_k when another hypothesis is true, is not zero since the h w-test statistics are based on a redundancy of only $h - l$. The different w-test statistics are correlated and a large correlation will imply a risk of identifying the error incorrectly. More complex data snooping procedures than the one described here can be adopted according to the application needs.

3. ***Adaptation***. Finally, the adaptation accounts for the identified error to avoid a biased solution accepting the null hypothesis. Hence, the identified alternative hypothesis H_{a_j} will become the new null hypothesis H_o computing, after calling $\mathbf{K} = \left(\mathbf{H}_i^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{H}_i\right)^{-1}\mathbf{H}_i^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}$,

$$\boldsymbol{\Sigma}_{\widehat{\nabla}\widehat{\nabla}} = \left(\mathbf{c}_k^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\boldsymbol{\Sigma}_{\hat{e}\hat{e}}^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{c}_k\right)^{-1} \tag{55}$$

$$\underline{\widehat{\nabla}}_i = \boldsymbol{\Sigma}_{\widehat{\nabla}\widehat{\nabla}}\mathbf{c}_k^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\underline{\hat{\mathbf{e}}}_{u,i} \tag{56}$$

$$\delta\hat{\mathbf{x}}_{u,a,i} = \mathbf{K}\left(\delta\underline{\tilde{\mathbf{y}}}_{u,i} - \mathbf{c}_k\underline{\widehat{\nabla}}_i\right) = \delta\hat{\mathbf{x}}_{u,i} - \left(\mathbf{H}_i^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{H}_i\right)^{-1}\mathbf{H}_i^T\boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}\mathbf{c}_k\underline{\widehat{\nabla}}_i \tag{57}$$

$$\boldsymbol{\Sigma}_{\delta\hat{\boldsymbol{x}}^a} = \boldsymbol{\Sigma}_{\delta\hat{\boldsymbol{x}}} + \mathbf{K}\mathbf{c}_k\boldsymbol{\Sigma}_{\nabla\nabla}\mathbf{c}_k^T\mathbf{K}^T \tag{58}$$

$$\delta\underline{\hat{\mathbf{y}}}_{u,a,i} = \mathbf{H}_i\delta\hat{\mathbf{x}}^a \quad \underline{\hat{\mathbf{e}}}_{u,a,i} = \delta\underline{\tilde{\mathbf{y}}}_{u,i} - \delta\underline{\hat{\mathbf{y}}}_{u,a,i} \quad \boldsymbol{\lambda} = \underline{\widehat{\nabla}}_i^T\boldsymbol{\Sigma}_{\widehat{\nabla}\widehat{\nabla}}^{-1}\underline{\widehat{\nabla}}_i \tag{59}$$

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As the data set may contain multiple errors, after a cycle of DIA, the procedure is to be started over, now with the new null hypothesis, until the Overall Model test in the detection step is no longer rejected, or until there is no redundancy. In the latter case, the null hypothesis gets automatically accepted (as $h = l$, $\delta\underline{\tilde{\mathbf{y}}}_{u,i} \in \mathcal{R}(\mathbf{H}_i)$ and $\underline{\hat{\mathbf{e}}}_{u,i} = 0$), but the reliability is very poor. Also the precision of the estimators for $\delta\hat{\mathbf{x}}$ has become poorer than with the original H_0, as adding more parameters the model has become weaker. In case the same measurement is identified in a subsequent iteration, the user can decide whether to exclude the satellite (i.e. its full set of measurements) for the current epoch or to adapt the solution for a measurement of a different signal from the same measurement, e.g. the one having the highest w-test value after the identified one, and continue with DIA cycle until the conditions mentioned above are verified.

3.2.7. Cycle slip detection and correction

Cycle slip detection and correction algorithm is derived by the approaches described in [13] and [14]. It is anyhow recommended to avoid the reparation of CS, performing the sole detection and reinitialising the corresponding satellite filter parameters.

The method is based on the implementation of the following steps for each epoch k and for each tracked satellite s. The numerical values of CS related parameters, like thresholds or window lengths, are provided in Appendix F.

1) Declaration of CS if the data gap is greater than Δt_{MAX}.
2) Application to carrier phase measurements of CS detected at previous epochs if any.
3) Computation of Melbourne-Wübbena (MW) combination as

$$\begin{aligned}\underline{\Phi}^s_{u,\mathrm{mw}_j} &= \underline{\Phi}^s_{u,\mathrm{wl}_j} - \underline{\mathrm{P}}^s_{u,\mathrm{nl}_j} = \left(\frac{f_1}{f_1 - f_j}\underline{\Phi}^s_{u,1} - \frac{f_j}{f_1 - f_j}\underline{\Phi}^s_{u,j}\right) - \left(\frac{f_1}{f_1 + f_j}\underline{P}^s_{u,1} + \frac{f_j}{f_1 + f_j}\underline{P}^s_{u,j}\right) \\ &= \lambda_{\mathrm{wl}_j}\left(M^s_{u,1} - M^s_{u,j}\right) + b_{u,nl_j} - b^s_{nl_j} + \beta_{u,wl_j} - \beta^s_{nwl_j} + W^s_{u,\mathrm{wl}_j} + \underline{\epsilon}^s_{u,\mathrm{mw}_j}\end{aligned} \tag{60}$$

where $\lambda_{\mathrm{wl}_j} = \frac{c}{f_1 - f_j} = \frac{\lambda_1\lambda_j}{\lambda_j - \lambda_1}$, WL indicates the Wide-Lane combination $(\cdot)^{\cdot}_{,\mathrm{wl}_j} = \frac{f_1}{f_1 - f_j}(\cdot)^{\cdot}_{,1} - \frac{f_j}{f_1 - f_j}(\cdot)^{\cdot}_{,j}$, NL the Narrow-Lane one $(\cdot)^{\cdot}_{,nl_j} = \frac{f_1}{f_1 + f_j}(\cdot)^{\cdot}_{,1} + \frac{f_j}{f_1 + f_j}(\cdot)^{\cdot}_{,j}$ and $\underline{\epsilon}^s_{u,\mathrm{mw}_j} = \underline{\epsilon}^s_{u,\mathrm{nl}_j} + \underline{\varepsilon}^s_{u,\mathrm{wl}_j}$.

4) Computation of the geometry-free combination of the phase measurements as

$$\begin{aligned}\underline{\Phi}^s_{u,\mathrm{gf}_j} &= \underline{\Phi}^s_{u,1} - \underline{\Phi}^s_{u,j} \\ &= -\left(1 - \mu_j\right)I^s_u + \beta_{u,\mathrm{gf}_j} - \beta^s_{\mathrm{gf}_j} + W^s_{u,\mathrm{gf}_j} + M^s_{u,\mathrm{gf}_j} + \underline{\varepsilon}^s_{u,\mathrm{gf}_j}\end{aligned} \tag{61}$$

where $M^s_{u,\mathrm{gf}_j} = \lambda_1 M^s_{u,1} - \lambda_j M^s_{u,j}$ and in general $(\cdot)^{\cdot}_{,\mathrm{gf}_j} = (\cdot)^{\cdot}_{,1} - (\cdot)^{\cdot}_{,j}$.

5) Computation of the integrated Doppler at the current epoch, $\underline{\Phi}^s_{u,\mathrm{di}_m}(t_k)$, as

$$\underline{\Phi}^s_{u,\mathrm{di}_m}(t_k) = \underline{\Phi}^s_{u,m}(t_{k-1}) + \Delta t\,\frac{\underline{\mathrm{D}}^s_{u,m}(t_k) + \underline{\mathrm{D}}^s_{u,m}(t_{k-1})}{2},\ \Delta t = t_k - t_{k-1} \tag{62}$$

6) Computation of the difference

$$\delta\overline{\underline{\Phi}}^s_{u,\mathrm{mw}_j} = \left|\underline{\Phi}^s_{u,\mathrm{mw}_j} - \overline{\Phi}^s_{u,\mathrm{mw}_j}(k-1, k-1-l_{\mathrm{mw}})\right| \tag{63}$$

between the MW combination $\underline{\Phi}^s_{u,\mathrm{mw}_j}$ for the current epoch k and the mean $\overline{\Phi}^s_{u,\mathrm{wl}_j}(k-1, k-1-l_{\mathrm{mw}})$ of the MW combinations over the measurements of the l_{mw} previous epochs.

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7) Computation of the time differences of the MW combinations between current and previous epochs as

$$\Delta\underline{\Phi}^{s}_{u,\mathrm{mw}_j} = \left|\underline{\Phi}^{s}_{u,\mathrm{mw}_j}(t_k) - \underline{\Phi}^{s}_{u,\mathrm{mw}_j}(t_{k-1})\right| \tag{64}$$

between the MW combination $\underline{\Phi}^{s}_{u,\mathrm{mw}_j}$ for the current epoch k and the mean $\overline{\Phi}^{s}_{u,\mathrm{wl}_j}(k-1, k-1-l_{\mathrm{mw}})$ of the MW combinations over the measurements of the l_{mw} previous epochs.

8) Computation of the time differences of the geometry-free combinations between current and previous epochs as

$$\underline{\Delta\Phi}^{s}_{u,\mathrm{gf}_j} = \underline{\Phi}^{s}_{\mathrm{u,gf}_j}(t_k) - \underline{\Phi}^{s}_{\mathrm{u,gf}_j}(t_{k-1}) \text{ for } j = 2, \dots, M. \tag{65}$$

9) Fitting of second order polynomials $P_{\mathrm{gf_j}}(s, l_{\mathrm{gf}})$ to the previous $l_{\mathrm{gf}} < L_{gf}$ values (after the last cycle slip and if $l_{\mathrm{gf}} > 2$) $\underline{\Phi}^{s}_{u,\mathrm{gf}_j}(\mathrm{k} - l_{\mathrm{gf}}), \dots, \underline{\Phi}^{s}_{u,\mathrm{gf}_j}(k-1)$, for $j = 1, \dots, M_{\Sigma(\mathrm{s})}$.

10) Computation of the differences, for $j = 2, \dots, M_{\Sigma(\mathrm{s})}$

$$\delta\underline{\Phi}^{s}_{u,\mathrm{gf}_j} = \left|\underline{\Phi}^{s}_{u,\mathrm{cs}_j} - P_{\mathrm{gf}_j}(s, t_k)\right| \tag{66}$$

between $\underline{\Phi}^{s}_{u,\mathrm{gf}_j}$ at the current epoch t_k and the prediction of $P_{\mathrm{gf}_j}(s, l_{\mathrm{gf}})$ at epoch t_k, $P_{\mathrm{gf}_j}(s, t_k)$.

11) Computation of the ionospheric rate

$$\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k) = \frac{\underline{\Delta\Phi}^{s}_{u,\mathrm{gf}_j}(t_k, t_{k-1})}{(\mu_j - 1)\Delta t} \text{ for } j = 2, \dots, M. \tag{67}$$

12) If the ionospheric rates of previous epochs $\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-1})$ and $\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-2})$ are set, computation of the predicted ionospheric rate

$$\hat{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k) = \dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-1}) + (t_k - t_{k-1})\frac{\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-1}) - \dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-2})}{\mathrm{t_{k-1} - t_{k-2}}} \text{ for } j = 2, \dots, M. \tag{68}$$

13) If the predicted ionospheric rate is available, computation of the differences, for $j = 2, \dots, M_{\Sigma(\mathrm{s})}$

$$\delta\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j} = \left|\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k) - \hat{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf_j}}(t_k)\right| \tag{69}$$

between the ionospheric rate computed at the current epoch, $\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k)$, and the predicted one from the previous CS-free epochs, $\hat{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k)$.

14) Computation of the differences, for $m = 1, \dots, M_{\Sigma(\mathrm{s})}$,

$$\delta\underline{\Phi}^{s}_{u,\mathrm{di}_m} = \left|\underline{\Phi}^{s}_{u,\mathrm{di}_m}(t_k) - \underline{\Phi}^{s}_{u,m}(t_k)\right| \tag{70}$$

between the Doppler integrations and the carrier phase measurements at the current epoch.

15) Declaration of a CS if

$$\begin{aligned}&\left(\left(\delta\overline{\Phi}^{s}_{u,\mathrm{mw}_j} > \epsilon_{\mathrm{mw}}\right) \wedge \left(\delta\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j} > \epsilon_{\mathrm{ir}} \vee \delta\underline{\Phi}^{s}_{u,\mathrm{gf}_j} > \epsilon_{\mathrm{gf}} \vee \Delta\underline{\Phi}^{s}_{u,\mathrm{gf}_j} > \epsilon_{\mathrm{gf}}\right)\right) \vee \\ &\left(\delta\underline{\Phi}^{s}_{u,\mathrm{di}_m} > k_{\mathrm{di}} \wedge \left|\mathrm{nint}\left(\delta\underline{\Phi}^{s}_{u,\mathrm{di}_m}\right)\right| < f_{\mathrm{di}}\right) \vee \mathrm{LLi_{rx}}\end{aligned} \tag{71}$$

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where

- ∧ and ∨ are respectively the logical operators AND and OR,
- nint indicates the nearest integer function,
- $\epsilon_{\mathrm{mw}} = k_{\mathrm{mw}}\sigma_{\overline{\Phi}_{\mathrm{mw}_j}}$,
- $\epsilon_{\mathrm{ir}} = a_{0,\mathrm{ir}} - a_{0,\mathrm{ir}}\alpha_{\mathrm{ir}}e^{-\Delta t/T}$,
- $\epsilon_{\mathrm{gf}} = a_{0,\mathrm{gf}} - a_{0,\mathrm{gf}}\alpha_{\mathrm{gf}}e^{-\Delta t/T}$,
- k_{mw}, $a_{0,\mathrm{ir}}$, α_{ir}, $a_{0,\mathrm{gf}}$, α_{gf}, T, k_{di} and f_{di} are user set threshold parameters (see Appendix F),
- and $\sigma_{\overline{\Phi}_{\mathrm{mw}_j}}$ is the standard deviation of the MW combinations over the measurements of the previous l_{mw} epochs.

$$\overline{\underline{\Phi}}^{s}_{u,\mathrm{wl}_j}(\mathrm{k},\mathrm{k}-\mathrm{l}_{\mathrm{mw}}) = \frac{m-1}{m}\overline{\underline{\Phi}}^{s}_{u,\mathrm{wl}_j}(k-1,k-1-l_{\mathrm{mw}}) + \frac{1}{m}\underline{\Phi}^{s}_{u,\mathrm{mw}_j} \tag{72}$$

$$\sigma_{\overline{\Phi}_{\mathrm{mw}}}(\sigma^2_{\overline{\Phi}_{\mathrm{mw}}}(k,k-l_{\mathrm{mw}}) = \frac{m-1}{m}\sigma_{\overline{\Phi}_{\mathrm{mw}_j}}(\sigma^2_{\overline{\Phi}_{\mathrm{mw}_j}}(k-1,k-1-l_{\mathrm{mw}}) + \frac{1}{m}\left(\Delta\overline{\underline{\Phi}}^{s}_{u,\mathrm{mw}_j}\right)^2 \tag{73}$$

16) If CS is detected compute the CS magnitude as

$$\Delta N^{s}_{u,j} = \mathrm{nint}\left(\frac{\lambda_1}{\lambda_j-\lambda_1}\left(\mathrm{nint}\left(\frac{\Delta\underline{\Phi}^{s}_{u,\mathrm{mw}_j}}{\lambda_{\mathrm{mw}}}\right) + \frac{\hat{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k)(\mu_j-1)\Delta t - \Delta\underline{\Phi}^{s}_{u,\mathrm{gf}_j}}{\lambda_1}\right)\right), \text{for } j = 2,\dots,M_{\Sigma(\mathrm{s})} \tag{74}$$

$$\Delta N^{s}_{u,1} = \mathrm{nint}\left(\frac{\Delta\underline{\Phi}^{s}_{u,\mathrm{mw}_2}}{\lambda_{\mathrm{mw}}}\right) - \Delta N^{s}_{u,2} \tag{75}$$

with nint indicating the nearest integer function, and repair the phase measurements as

$$\underline{\Phi}^{s}_{u,m} = \underline{\Phi}^{s}_{u,m} - \lambda_m\Delta N^{s}_{u,m}, \text{ for } m = 1,\dots,M_{\Sigma(\mathrm{s})} \tag{76}$$

Repeat items from 3 to 15. If no CS is further detected the CS is marked as repaired, the modification of the phase measurement maintained and the CS magnitudes $\Delta N^{s}_{u,m}$ stored for correcting the carrier phase measurements at the following epochs. Otherwise, the CS are marked as not repaired and the modifications to the carrier phase measurements are not kept.

17) If no CS was detected or it was repaired:

- Update of the mean $\overline{\phi}^{s}_{u,\mathrm{wl}_j}$ and the standard deviation $\sigma_{\overline{\Phi}_{\mathrm{mw}_j}}$ of the MW combinations over the measurements of the previous l_{mw} epochs with the values for the last epoch as

$$\overline{\underline{\Phi}}^{s}_{u,\mathrm{wl}_j}(\mathrm{k},\mathrm{k}-\mathrm{l}_{\mathrm{mw}}) = \frac{m-1}{m}\overline{\underline{\Phi}}^{s}_{u,\mathrm{wl}_j}(k-1,k-1-l_{\mathrm{mw}}) + \frac{1}{m}\underline{\Phi}^{s}_{u,\mathrm{mw}_j} \tag{77}$$

$$\sigma^2_{\overline{\Phi}_{\mathrm{mw}}}(k,k-l_{\mathrm{mw}}) = \frac{m-1}{m}\sigma^2_{\overline{\Phi}_{\mathrm{mw}_j}}(k-1,k-1-l_{\mathrm{mw}}) + \frac{1}{m}\left(\Delta\overline{\underline{\Phi}}^{s}_{u,\mathrm{mw}_j}\right)^2 \tag{78}$$

  with $\sigma_{\overline{\Phi}_{\mathrm{mw}_j}}$ initialised as $\lambda_{\mathrm{wl}_j}/2$.

- Store the ionospheric rates as $\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-2}) = \dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-1})$, $\dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_{k-1}) = \dot{\underline{\mathrm{I}}}^{s}_{u,\mathrm{gf}_j}(t_k)$.
- Update the values of the geometry-free combinations $\underline{\Phi}^{s}_{u,\mathrm{gf}_j}$ for the polynomial fitting.

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18) If a CS was unsuccessfully repaired, reinitialise $\sigma_{\overline{\Phi}_{\mathrm{mw}_j}} = \lambda_{\mathrm{wl}_j}/2$ and $\overline{\underline{\Phi}}^{s}_{u,\mathrm{mwl}_j} = \underline{\Phi}^{s}_{\mathrm{u,mw_j}}$ and reset the stored geometry-free combination and ionospheric rate values. The satellites for which the CS repairing was unsuccessful are marked as rising satellites and their ionosphere and ambiguity parameters in the EKF state vector reinitialised as explained in Section 3.3.5.

3.2.8. Reference satellite selection

As introduced in subsection 2.1, the proposed PPP KF processes satellite-satellite SD of corrected measurements between any satellite and a reference satellite within each constellation. For each constellation, the reference satellite is selected as the satellite with the highest pivot selection factor $w^{s}_{\mathrm{psf},u}$ among the ones with the following characteristics:

- available carrier phase, pseudorange and Doppler measurements.
- measurements with sufficient quality (i.e. continuous carrier phase lock and valid pseudorange measurements: the lock time should be at least equal to 10 seconds);
- flagged as healthy and in the nominal orbit (i.e. eccentricity of the orbit *e* <0.01);
- available valid HAS corrections.

The pivot selection criteria is finding the maximum among the weighted average of three min-max normalised factors computed per satellite: the normalised pivot precedence, the normalised tracking period and the normalised cycle slips count factors. The pivot precedence factor is a function of the satellite s elevation sinusoidal mapping function $m^{s}_{P,u}$ and the average CN0 of its tracked signals defined as

$$w^{s}_{\mathrm{ppf},u} = \frac{1}{m^{s}_{P,u}\left(\frac{CN0_{min}}{CN0^{s}_{u}}\right)^{\mathrm{k}}} =, \text{for } m = 1, \dots, M_{\Sigma(\mathrm{s})} \tag{79}$$

The pivot precedence factor is then normalised with a min-max normalisation to provide the normalised pivot precedence factor $\overline{w}^{s}_{\mathrm{psf},u}$. The normalised tracking period factor, $\overline{w}^{s}_{\mathrm{tpf},u}$ is the min-max normalisation of the satellite tracking periods, i.e. the time interval since the receiver is tracking the signals of the satellites in view without interruptions. The normalised cycle slips count factor is the inverse min-max normalisation of the number of cycle slips detected, $\overline{w}^{s}_{\mathrm{csf},u}$ since the beginning of the tracking. The pivot score value is then computed as

$$p_S = \frac{1}{2}\overline{w}^{s}_{\mathrm{psf},u} + \frac{1}{4}\overline{w}^{s}_{\mathrm{tpf},u} + \frac{1}{4}\overline{w}^{s}_{\mathrm{csf},u}. \tag{80}$$

At each epoch, for each constellation, the satellite that has the maximum number of measurements, all the HAS corrections, the highest pivot score and supports the pivot criteria is selected as pivot.

3.3. NAV Estimation

The estimation of the NAV solution is provided by a real-time PPP algorithm that exploits HAS corrected satellite-satellite SD measurements, detailed in Section 3.3.3. As mentioned in Section 2.1, the current HAS-UA is based on a PPP with a float ambiguity solution The PPP is based on a EKF estimator whose implementation details are reported in the following sections.

The EKF is a recursive estimator allowing estimating the state of a linearized system disturbed by white noise from measurements linearly dependent on the state and corrupted by white noise. For this purpose, the EKF combines the available and the a priori knowledge of the system and measurements to estimate the unknown parameters by minimizing the mean square error. Details on EKF theory can be found in [15] while in the following an overview of the main steps necessary for the EKF implementation is given to introduce

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the next sections to the reader. In a discrete KF, the state equation and measurement equation may be written as:

$$\overline{\underline{\mathbf{x}}}_{u,k} = \mathbf{\Phi}_{k,k-1}\overline{\underline{\mathbf{x}}}_{u,k-1} + \underline{\mathbf{w}}_{u,k} \tag{81}$$

$$\underline{\tilde{\mathbf{z}}}_{u,k} = \mathbf{H}_k\overline{\underline{\mathbf{x}}}_{u,k} + \underline{\mathbf{v}}_{u,k} \tag{82}$$

where

- $\overline{\underline{\mathbf{x}}}_{u,k} = \overline{\underline{\mathbf{x}}}_k$ is the state vector, at the instant k, composed of the estimable unknown parameters and $\overline{\underline{\mathbf{x}}}_{\mathrm{u},k-1}$ is the state vector at the previous instant $k-1$,
- $\mathbf{\Phi}_{k,k-1}$ is the transition matrix relating the state at the previous instant $k-1$ to the one of the current instant k,
- $\underline{\tilde{\mathbf{z}}}_{u,k} = \underline{\tilde{\mathbf{z}}}_k$ is the measurement vector,
- $\mathbf{H}_k$ is the design matrix which relates the state vector to the measurements,
- $\underline{\mathbf{w}}_{u,k} = \underline{\mathbf{w}}_k \sim \mathrm{N}(0, \mathbf{\Sigma}_{\mathrm{ww}})$ and $\underline{\mathbf{v}}_{u,k} = \underline{\mathbf{v}}_k \sim \mathrm{N}(0, \mathbf{\Sigma}_{\mathrm{vv}})$ are respectively the process and measurement noise vectors assumed independent and with a normal probability distribution.

The implementation of the EKF requires two iterative steps: the prediction update, described in Section 3.3.2, and the measurement update, described in Section 3.3.8.

3.3.1. Initialization of NAV receiver parameters from PP SE PVT

The initialization of the NAV solution consists of the initialization of the receiver related parameters in the NAV state vectors (receiver position, velocity, acceleration and wet troposphere delay) and their covariance matrices. All the observed satellites are flagged as newly tracked (rising), so that their ionosphere and ambiguity parameters must be initialised in Step 11.

The position and velocity in the NAV state vector are initialized using the SE PVT solution for position and velocity, $\underline{\hat{\boldsymbol{r}}}_u, \underline{\hat{\boldsymbol{v}}}_u$ together with their covariance matrices $\mathbf{\Sigma}_{\hat{\mathbf{r}}_u}$ and $\mathbf{\Sigma}_{\hat{\mathbf{v}}_u}$. The receiver acceleration $\tilde{\mathbf{a}}_u$ in the NAV state vector can (depending on the application) either be initialised to zero or obtained as difference from the velocities estimated in two subsequent $\underline{\hat{\boldsymbol{v}}}_u$ SE PVT estimations divided by the time interval between them, while its variance can also be initialised from the SE PVT velocity variances or set to an a priori value $\mathbf{\Sigma}_{\tilde{\mathbf{a}}_u}$. The residual wet zenith troposphere delay can be initialised to zero, $\tilde{T}_{\delta \mathrm{w},u} = 0$ and its variance to an a priori value $\sigma^2_{\tilde{T}_{\delta \mathrm{W},u}}$. The ionosphere slant delays and the ambiguities are initialised as shown in Section 3.3.5 for the rising satellites, obtaining the parameters $\underline{\hat{\boldsymbol{I}}}_u^{Sp}$ *and* $\underline{\hat{\boldsymbol{N}}}_{u,M}^{Sp}$ and their variances $\mathbf{\Sigma}_{\tilde{\mathbf{I}}_u^S}$ and $\mathbf{\Sigma}_{\tilde{N}_{u,M}^S}$. This section describes also how to set these initial parameters, even if they are derived only in Step 11. The above listed a priori information composes the following vector:

$$\underline{\hat{\mathbf{x}}}_0 = \left[\underline{\hat{\boldsymbol{r}}}_u, \underline{\hat{\boldsymbol{v}}}_u, \tilde{\mathbf{a}}_u, \tilde{T}_{\delta \mathrm{w},u}, \underline{\hat{\boldsymbol{I}}}_u^{Sp^T}, \underline{\hat{\boldsymbol{N}}}_{u,M}^{Sp\ T}\right]^{\mathrm{T}} \tag{83}$$

The initial state vector covariance matrix is obtained combining the variances mentioned above as

$$\mathbf{\Sigma}_{\hat{\mathbf{x}}_0} = \begin{bmatrix} \mathbf{\Sigma}_{\hat{\mathbf{r}}_u} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{\Sigma}_{\hat{\mathbf{v}}_u} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{\Sigma}_{\tilde{\mathbf{a}}_u} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{\tilde{T}_{\delta \mathrm{W},u}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{\Sigma}_{\tilde{\mathbf{I}}_u^S} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{\Sigma}_{\tilde{N}_{u,M}^S} \end{bmatrix} \tag{84}$$

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where $\mathbf{\Sigma}_{\bar{I}_u}$ and $\mathbf{\Sigma}_{\bar{N}_u}$ are defined as

$$\mathbf{\Sigma}_{\bar{\mathbf{I}}_u^S} = \oplus_{\Sigma=\mathrm{E,G}} \mathbf{D}_\Sigma \begin{bmatrix} \sigma_{\bar{I}_u^1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_{\bar{I}_u^{S_\Sigma}} \end{bmatrix} \mathbf{D}_\Sigma^T \tag{85}$$

$$\mathbf{\Sigma}_{\bar{N}_{u,M}^S} = \oplus_{\Sigma=\mathrm{E,G}} \mathbf{1}_{M_\Sigma} \otimes \left[\mathbf{D}_\Sigma \begin{pmatrix} \sigma_{\bar{N}_u^1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_{\bar{N}_u^{S_\Sigma}} \end{pmatrix} \mathbf{D}_\Sigma^T \right] \tag{86}$$

3.3.2. Linear prediction of the state estimate

The prediction process allows obtaining an a priori propagation of the previous state and its error covariance for the current instant**.** This step is performed alternatively to the receiver parameter initialisation. In a discrete KF, the state equation and measurement equation may be written as:

$$\breve{\underline{\mathbf{x}}}_k = \mathbf{\Phi}_{k,k-1} \hat{\underline{\mathbf{x}}}_{u,k-1} \tag{87}$$

$$\mathbf{\Sigma}_{\breve{\mathbf{x}}_k} = \mathbf{\Phi}_{k,k-1} \mathbf{\Sigma}_{\hat{\mathbf{x}}_{k-1}} \mathbf{\Phi}_{k,k-1}^T + \Delta t \cdot \mathbf{\Sigma}_{w_k} \tag{88}$$

where

- $\breve{\underline{\mathbf{x}}}_k$ is the prediction at the instant t_k of the state vector,
- $\hat{\underline{\mathbf{x}}}_{k-1}$ is the estimation of the state vector at the previous instant t_{k-1},
- $\mathbf{\Phi}_{k,k-1}$ is the transition matrix relating the state at the previous instant t_{k-1} to the one of the current instant t_k,
- $\mathbf{\Sigma}_{\breve{\mathbf{x}}_k}$ is the covariance matrix of the predicted state vector,
- $\mathbf{\Sigma}_{\hat{\mathbf{x}}_{k-1}}$ is the covariance matrix of the estimated state vector at the previous epoch,
- $\mathbf{\Sigma}_{w_k}$ is the covariance matrix of the system noise (or process noise),
- Δt is the time interval between two subsequent measurements.

The state vector of the proposed NAV EKF for a specific instant k is defined as

$$\overline{\mathbf{x}}_k = \left(\overline{\mathbf{r}}_u^T, \overline{\mathbf{v}}_u^T, \overline{\mathbf{a}}_u^T, \overline{T}_{\delta \mathrm{w},u}, \bar{\mathbf{i}}_u^{Sp^T}, \overline{\mathbf{n}}_{u,M}^{Sp\ T} \right)^{\mathrm{T}} \tag{89}$$

where

- $\bar{\mathbf{r}}_u$ is the vector of the user position,
- $\bar{\mathbf{v}}_u$ is the vector of the user velocity,
- $\bar{\mathbf{a}}_u$ is the vector of the user acceleration,
- $\overline{T}_{\delta \mathrm{w},u}$ is the residual zenith wet tropospheric delay,
- $\bar{\mathbf{i}}_u^{Sp} = \left[I_u^{1p_E}, \ldots, I_u^{(S_E-1)p_E}, I_u^{1p_E}, \ldots, I_u^{(S_G-1)p_G} \right]^T$ is the vector of the SD ionospheric slant delays composed of $(S_E + S_G - 2)$ elements, with $S = \mathrm{S_E} + S_G$ indicating the total number of Galileo and GPS tracked satellites that includes the reference satellites, and p_E and p_G indicating the Galileo and GPS reference (or pivot) satellites respectively,
- $\bar{\mathbf{n}}_{u,M}^{Sp} = \left[N_{u,1}^{1p_E}, \ldots, N_{u,1}^{(S_E-1)p_E}, \ldots, N_{u,M_E}^{1p_E}, \ldots, N_{u,M_E}^{(S_E-1)p_E}, N_{u,1}^{1p_G}, \ldots, N_{u,1}^{(S_G-1)p_G}, \ldots, N_{u,M_G}^{1p_G}, \ldots, N_{u,M_G}^{(S_G-1)p_G} \right]^T$ is the SD carrier phase ambiguity vector composed of $(S - \mathrm{C})M$ elements, with C the number of adopted

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GNSS constellations (1 Galileo only, 2 Galileo+GPS). In the following sections, details on the HAS-UA KF implementation are provided.

A constant-acceleration model relates the position, velocity and acceleration parameters at epochs t_k and t_{k-1}, namely

$$\begin{aligned}\overline{\mathbf{r}}_k &= \overline{\mathbf{r}}_{k-1} + \Delta t\overline{\mathbf{v}}_{k-1} + \frac{\Delta t^2}{2}\overline{\mathbf{a}}_{k-1}\\ \overline{\mathbf{v}}_k &= \overline{\mathbf{v}}_{k-1} + \Delta t\overline{\mathbf{a}}_{k-1}\\ \overline{\mathbf{a}}_k &= \overline{\mathbf{a}}_{k-1}\end{aligned} \tag{90}$$

The state prediction is based on the state space model, which uses the state estimate $\underline{\hat{\mathbf{x}}}_{k-1}$ of the previous epoch $k-1$ (at the first epoch, the initial state described in Section 3.3.1 is used) and the state transition matrix $\mathbf{\Phi}_{k,k-1}$ to obtain the predicted state vector given by Eq. (87). The state transition matrix $\mathbf{\Phi}_{k,k-1}$ is defined as

$$\mathbf{\Phi}_{k,k-1} = \begin{pmatrix} \mathbf{I}_3 & \Delta t\cdot\mathbf{I}_3 & \frac{\Delta t^2}{2}\cdot\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}_3 & \Delta t\cdot\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 1 & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{\mathrm{S-C}} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{\mathbf{A}} \end{pmatrix} \tag{91}$$

where $\mathrm{A} = \mathrm{M_E}(\mathrm{S_E}-1) + \mathrm{M_G}(\mathrm{S_G}-1)$ is the number of estimable ambiguities. It should be underlined that when the reference satellite changes it is not necessary to re-initialize all SD ionosphere slant delays and float ambiguities. These SD parameters can be simply propagated according to

$$\bar{N}_{u,m}^{sq} = \bar{N}_{u,m}^{sp} - \bar{N}_{u,m}^{qp} \tag{92}$$

$$\bar{I}_u^{sq} = \bar{I}_u^{sp} - \bar{I}_u^{qp} \tag{93}$$

with q and p indicating respectively the new and the old reference satellites. The other state parameters (position, velocity, acceleration and tropospheric zenith delay) are not affected by the single differencing and, therefore, do not need to be adjusted. When a change of reference satellite happens, therefore, the state transformation matrix becomes

$$\mathbf{\Phi}_{k,k-1} = \begin{pmatrix} \mathbf{I}_3 & \Delta t\cdot\mathbf{I}_3 & \frac{\Delta t^2}{2}\cdot\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}_3 & \Delta t\cdot\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 1 & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{D}_I & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{D}_A \end{pmatrix} \tag{94}$$

where

- $\mathbf{D}_I = \bigoplus_{\Sigma=E,G}\mathbf{D}_\Sigma$, $\mathbf{D}_\Sigma = \begin{bmatrix}\mathbf{I}_{q_\Sigma-1} & -\mathbf{1}_{S_\Sigma-1} & \mathbf{0}\\ \mathbf{0} & & \mathbf{I}_{S_\Sigma-(q_\Sigma+1)}\end{bmatrix}$,
- $\mathbf{D}_A = \bigoplus_{\Sigma=E,G}\mathbf{D}_{M_\Sigma}$, $\mathbf{D}_{M_\Sigma} = \mathbf{I}_{M_\Sigma}\otimes\mathbf{D}_\Sigma$.

Please note that measurements from the satellite used as pivot at the previous epoch, if present at the current epoch, should be treated as measurements from a rising satellite and the corresponding parameters in the state vector initialised as indicated in Section 3.3.5.

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3.3.3. Computation of EKF measurements and their covariance matrix

The measurements $\underline{\tilde{\mathbf{z}}}_k$ of the NAV filter consist of the HAS corrected satellite-satellite SD of carrier phase, pseudorange, and Doppler measurements for all the tracked satellites and selected signal frequencies. The SD process allows eliminating all common receiver errors, such as the clock offset, receiver code and phase bias, and the phase wind-up effect due to the receiver.

$$\begin{aligned}\underline{\Phi}_{u,m}^{sp} &= \underline{\Phi}_{u,m}^{s} - \underline{\Phi}_{u,m}^{p} = \\ &\|\mathbf{r}_u - \mathbf{r}^s\| - \|\mathbf{r}_u - \mathbf{r}^p\| - dt^{sp} + m_{H,u}^{sp} T_{H,u} + m_{W,u}^{sp} T_{W,u} - \mu_m I_u^{sp} - \lambda_m \beta_m^{sp} + \lambda_m M_{u,m,i-1}^{sp} + \delta t_{Sha,u}^{sp} \\ &\quad + \zeta_{\mathrm{PCO},u,m}^{sp} + \zeta_{\mathrm{PCV},u,m}^{sp} + \lambda_m W_m^{sp} + \underline{\varepsilon}_{u,m}^{sp}\end{aligned} \tag{95}$$

$$\begin{aligned}\underline{P}_{u,m}^{sp} &= \underline{\mathrm{P}}_{u,m}^{s} - \underline{\mathrm{P}}_{u,m}^{p} = \\ &\|\mathbf{r}_u - \mathbf{r}^s\| - \|\mathbf{r}_u - \mathbf{r}^p\| - dt^{sp} + m_{H,u}^{sp} T_{H,u} + m_{W,u}^{sp} T_{W,u} + \mu_m I_u^{sp} - \lambda_m b_m^{sp} + \zeta_{\mathrm{PCO},u,m}^{sp} + \delta t_{Sha,u}^{sp} + \underline{\epsilon}_{u,m}^{s}\end{aligned} \tag{96}$$

$$D_{u,m}^{sp} = \underline{D}_{u,m}^{s} - \underline{D}_{u,m}^{p} = \|\mathbf{v}_u - \mathbf{v}^s\| - \|\mathbf{v}_u - \mathbf{v}^p\| - \dot{dt}^{sp} + \underline{\Xi}_{u.m}^{sp} \tag{97}$$

where the upper indices p is used to indicate the measurements of the reference satellite selected as described in Section 3.2.7. In the rest of the document, the index p will be used to indicate a generic reference satellite, and therefore it could refer to both GPS and Galileo reference satellites, while p_e and p_g will indicate the reference Galileo and GPS satellites, respectively.

Differently from the SE PVT estimation, in the NAV filter the slant ionosphere delays are estimated as parameters in the state vector and no ionosphere-free combination is performed on the measurements. For the tropospheric correction, on top of correcting the measurement by a priori model corrections, also a residual wet zenith tropospheric delay is estimated in the state vector. The expression for the observed minus computed SD phase, pseudorange and Doppler measurements become therefore

$$\begin{aligned}\delta\underline{\tilde{\Phi}}_{u,m}^{sP} &= \underline{\Phi}_{u,m}^{sp} + \mathbf{e}_u^s \tilde{\mathbf{r}}^s - \mathbf{e}_u^p \tilde{\mathbf{r}}^p + \widetilde{dt}^{sp} - \widetilde{m}_{\mathrm{H},u}^{sp} \tilde{T}_{\mathrm{H},u} - \widetilde{m}_{\mathrm{W},u}^{sp} \tilde{T}_{\mathrm{W},u} + \lambda_m \tilde{\beta}_m^{sp} + \tilde{\zeta}_{\mathrm{PCO},u,m}^{sp} + \tilde{\zeta}_{\mathrm{PCV},u,m}^{sp} \\ &\quad + \lambda_m \widetilde{W}_{\mathrm{u,m}}^{sp} + \delta t_{Sha,u}^{sp} \\ &= \mathbf{e}_u^{sp} \bar{\mathbf{r}}_{u,i} + \widetilde{m}_{\mathrm{W},u}^{sp} \bar{T}_{\delta\mathrm{W},u,i} - \mu_m \bar{I}_{u,i}^{sp} + \lambda_m \bar{N}_{u,m,i}^{sp} + \underline{\varepsilon}_{u,m}^{sp}\end{aligned} \tag{98}$$

$$\begin{aligned}\delta\underline{\tilde{P}}_{u,m}^{sp} &= \underline{\mathrm{P}}_{u,m}^{sp} + \mathbf{e}_u^s \tilde{\mathbf{r}}^s + \mathbf{e}_u^p \tilde{\mathbf{r}}^p + \widetilde{dt}^{sp} - \widetilde{m}_{\mathrm{H},u}^{sp} \tilde{T}_{\mathrm{H},u} - \widetilde{m}_{\mathrm{W},u}^{sp} \tilde{T}_{\mathrm{W},u} + \tilde{b}_m^{sp} + \tilde{\zeta}_{\mathrm{PCO},u,m}^{sp} + \delta t_{Sha,u}^{sp} \\ &= \mathbf{e}_u^{sp} \bar{\mathbf{r}}_{u,i} + \widetilde{m}_{\mathrm{W},u}^{sp} \bar{T}_{\delta\mathrm{W},u,i} + \mu_m \bar{I}_{u,i}^{sp} + \underline{\epsilon}_{u,m}^{sp}\end{aligned} \tag{99}$$

$$\delta\underline{\tilde{D}}_{u,m}^{sp} = \underline{D}_{u,m}^{sp} + \mathbf{e}_u^s \tilde{\mathbf{v}}^s - \mathbf{e}_u^p \tilde{\mathbf{v}}^p + \widetilde{\dot{dt}}^{sp} = \mathbf{e}_u^s \bar{\mathbf{v}}_{u,i} + \underline{\Xi}_{u.m}^{sp} \tag{100}$$

By defining $\underline{\tilde{\mathbf{z}}}_k^p = \left[\delta\underline{\tilde{\boldsymbol{\Phi}}}_{u,M_E}^{Ep_e}, \delta\underline{\tilde{\boldsymbol{\Phi}}}_{u,M_G}^{Gp_g}, \delta\underline{\tilde{\mathbf{p}}}_{u,M_E}^{Ep_e}, \delta\underline{\tilde{\mathbf{p}}}_{u,M_G}^{Gp_g}, \delta\underline{\tilde{\mathbf{d}}}_{u,M_E}^{Ep_e}, \delta\underline{\tilde{\mathbf{d}}}_{u,M_G}^{Gp_g}\right]^T$ as the vector of the observed minus computed SD measurements and $\delta\underline{\widetilde{\boldsymbol{m}}}_{u,M}^{\Sigma p} = \left[\delta\underline{\widetilde{\boldsymbol{m}}}_{u,1}^{1p}, \dots, \delta\underline{\widetilde{\boldsymbol{m}}}_{u,1}^{S_\Sigma p}, \delta\underline{\widetilde{\boldsymbol{m}}}_{u,2}^{1p}, \dots, \delta\underline{\widetilde{\boldsymbol{m}}}_{u,2}^{S_\Sigma p} \dots, \delta\underline{\widetilde{\boldsymbol{m}}}_{u,M_\Sigma}^{1p}, \dots, \delta\underline{\widetilde{\boldsymbol{m}}}_{u,M_\Sigma}^{S_\Sigma p}\right]^T$, with $\Sigma = E \mid G$ and $\underline{\widetilde{\boldsymbol{m}}} = \underline{\tilde{\boldsymbol{\Phi}}} \mid \underline{\tilde{\mathbf{p}}} \mid \underline{\tilde{\mathbf{d}}}$, as the vectors of the NAV observed minus computed $\underline{\widetilde{\boldsymbol{m}}}$ measurements for system Σ, the observed minus computed single differences can be written in matrix form as

$$\underline{\tilde{\mathbf{z}}}_k^p = \mathbf{T}_{SD}\left[\underline{\boldsymbol{\Phi}}_{u,i}^T, \underline{\mathbf{p}}_{u,i}^T, \underline{\mathbf{d}}_{u,i}^T, \tilde{\mathbf{r}}^{sT}, \tilde{\mathbf{v}}^{sT}, \widetilde{\mathbf{dt}}^{sT}, \widetilde{\dot{\mathbf{dt}}}^{sT}, \tilde{\boldsymbol{\beta}}_M^{sT}, \tilde{\mathbf{b}}_M^{sT}, \tilde{T}_{\mathrm{H},u}, \bar{T}_{\mathrm{W},u}, \widetilde{\mathbf{w}}^s, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{\mathrm{PCO}}, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{\mathrm{PCV}}\right]^{\boldsymbol{T}} \tag{101}$$

$$\mathbf{T}_{SD} = \begin{bmatrix} \mathbf{D} & \mathbf{0} & \mathbf{0} & \mathbf{E}_u & \mathbf{0} & \mathbf{D} & \mathbf{0} & \boldsymbol{\Lambda} & \mathbf{0} & -\mathbf{m}_{\mathrm{H},u}^S & -\mathbf{m}_{\mathrm{W},u}^S & -\mathbf{B} & -\mathbf{B} & -\mathbf{B} \\ \mathbf{0} & \mathbf{D} & \mathbf{0} & \mathbf{E}_u & \mathbf{0} & \mathbf{D} & \mathbf{0} & \mathbf{0} & \mathbf{B} & -\mathbf{m}_{\mathrm{H},u}^S & -\mathbf{m}_{\mathrm{W},u}^S & \mathbf{0} & -\mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D} & \mathbf{0} & \mathbf{E}_u & \mathbf{0} & \mathbf{D} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{102}$$

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where

- $\underline{\boldsymbol{\Phi}}_{u,i} = \begin{bmatrix} \underline{\boldsymbol{\Phi}}_{u,M}^{E} \\ \underline{\boldsymbol{\Phi}}_{u,M}^{G} \end{bmatrix}$, $\underline{\mathbf{p}}_{u,i} = \begin{bmatrix} \mathbf{p}_{u,i}^{E} \\ \underline{\mathbf{p}}_{u,i}^{G} \end{bmatrix}$, $\underline{\mathbf{d}}_{u,i} = \begin{bmatrix} \mathbf{d}_{u,i}^{E} \\ \underline{\mathbf{d}}_{u,i}^{G} \end{bmatrix}$,
- $\underline{\boldsymbol{\Phi}}_{u,i}^{\Sigma} = \left[\underline{\Phi}_{u,1,i}^{1}, \ldots, \underline{\Phi}_{u,1,i}^{S_\Sigma}, \underline{\Phi}_{u,2,i}^{1}, \ldots, \underline{\Phi}_{u,2,i}^{S_\Sigma} \ldots, \underline{\Phi}_{u,\mathrm{M}_\Sigma,i}^{1}, \ldots, \underline{\Phi}_{u,\mathrm{M}_\Sigma,i}^{S_\Sigma}\right]^T$ is the vector of the carrier phase measurements for system Σ, with $\Sigma = E \mid G$,
- $\widetilde{\boldsymbol{\beta}}_{M}^{S} = \left[\widetilde{\boldsymbol{\beta}}^{E^T}, \widetilde{\boldsymbol{\beta}}^{G^T}\right]^T$, $\widetilde{\boldsymbol{\beta}}^{\Sigma} = \left[\tilde{\beta}_{1}^{1}, \ldots, \tilde{\beta}_{1}^{S_\Sigma}, \ldots, \tilde{\beta}_{M_\Sigma}^{1}, \ldots, \tilde{\beta}_{M_\Sigma}^{S_\Sigma}\right]^T$,
- $\widetilde{\mathbf{w}}^{S} = \left[\widetilde{\mathbf{w}}^{E^T}, \widetilde{\mathbf{w}}^{G^T}\right]^T$, $\widetilde{\mathbf{w}}^{\Sigma} = \left[\widetilde{W}_{1}^{1}, \ldots, W_{1}^{S_\Sigma}, \ldots, \widetilde{W}_{M_\Sigma}^{1}, \ldots, \widetilde{W}_{M_\Sigma}^{S_\Sigma}\right]^T$
- $\boldsymbol{\delta}\mathbf{r}_{\mathrm{PCV}} = \left[\widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV}^{E\ T}, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV}^{G\ T}\right]^T$, $\widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV}^{\Sigma} = \left[\widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV,1}^{1}, \ldots, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV,1}^{\mathrm{S}_\Sigma}, \ldots, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV,M_\Sigma}^{1}, \ldots, \widetilde{\boldsymbol{\delta}\mathbf{r}}_{PCV,M_\Sigma}^{\mathrm{S}_\Sigma}\right]^T$
- $\mathbf{E}_u = \bigoplus_{\Sigma=E,G} \mathbf{E}_u^{\Sigma}$, $\mathbf{E}_u^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} \bigoplus_{s=1}^{p_\Sigma - 1} \mathbf{e}_u^{s\ T} & -\mathbf{1}_{S_\Sigma - 1} \otimes \mathbf{e}_u^{p_\Sigma T} & \bigoplus_{s=p_\Sigma+1}^{S_\Sigma} \mathbf{e}_u^{s\ T} \end{bmatrix}$,
- $\mathbf{D} = \bigoplus_{\Sigma=E,G} \mathbf{D}_\Sigma$, $\mathbf{D}_\Sigma = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} \mathbf{I}_{p_\Sigma - 1} & -\mathbf{1}_{S_\Sigma - 1} & \mathbf{I}_{S_\Sigma - p_\Sigma} \end{bmatrix}$,
- $\boldsymbol{\Lambda} = \bigoplus_{\Sigma=E,G} \boldsymbol{\Lambda}_\Sigma$, $\boldsymbol{\Lambda}_\Sigma = \left(\bigoplus_{\mathrm{m}=1}^{M_\Sigma} \boldsymbol{\lambda}_m\right) \otimes \begin{bmatrix} \mathbf{I}_{p_\Sigma - 1} & -\mathbf{1}_{S_\Sigma - 1} & \mathbf{I}_{S_\Sigma - p_\Sigma} \end{bmatrix}$,
- $\mathbf{B} = \bigoplus_{\Sigma=E,G} \mathbf{B}_\Sigma$, $\mathbf{B}_\Sigma = \mathbf{I}_{M_\Sigma} \otimes \begin{bmatrix} \mathbf{I}_{p_\Sigma - 1} & -\mathbf{1}_{S_\Sigma - 1} & \mathbf{I}_{S_\Sigma - p_\Sigma} \end{bmatrix}$,
- $\mathbf{m}_{\ni,u}^{S} = \begin{bmatrix} \mathbf{m}_{\ni,\mathrm{u}}^{E} \\ \mathbf{m}_{\ni,\mathrm{u}}^{G} \end{bmatrix}$, for ∋= H | W, $\mathbf{m}_{\ni,u}^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} \bigoplus_{s=1}^{p_\Sigma - 1} m_{\ni,\mathrm{u}}^{s} & -\mathbf{1}_{S_\Sigma - 1} \otimes m_{\ni,\mathrm{u}}^{p_\Sigma} & \bigoplus_{s=p_\Sigma+1}^{S_\Sigma} m_{\ni,\mathrm{u}}^{s} \end{bmatrix}$.

As a sample case, the matrix $\mathbf{D}$ in case of $S = 5$ and $p = 4$ for system Σ, is given by the following matrix

$$\mathbf{D}_\Sigma = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \tag{103}$$

For this sample case, in the above matrix, the fourth column corresponds to the reference satellite ($p = 4$). For a general case, the column corresponding to the reference satellite should be filled with -1 values to perform the difference operation. The above matrix should be recomputed every time the reference satellite changes.

If the full $\boldsymbol{\Sigma}_{cc}$ matrix is available, the covariance matrix of the observed minus computed measurements $\boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k}$ is computed through the differencing operator $\mathbf{T}_{\mathrm{SD}}$ as

$$\boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k} = \mathbf{T}_{\mathrm{SD}} \begin{bmatrix} \boldsymbol{\Sigma}_{\overline{yy}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{cc} \end{bmatrix} \mathbf{T}_{\mathrm{SD}}^{\ T} \tag{104}$$

where

- $\boldsymbol{\Sigma}_{\overline{yy}} = \mathrm{diag}\left(\boldsymbol{\Sigma}_{\overline{\Phi\Phi}}^{E}, \boldsymbol{\Sigma}_{\overline{\Phi\Phi}}^{G}, \boldsymbol{\Sigma}_{\overline{PP}}^{E}, \boldsymbol{\Sigma}_{\overline{PP}}^{G}, \boldsymbol{\Sigma}_{\overline{DD}}^{E}, \boldsymbol{\Sigma}_{\overline{DD}}^{G}\right)$.
- $\boldsymbol{\Sigma}_{\overline{\Phi\Phi}}^{\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} {\sigma_{\Phi,u,m}^{s}}^2$ is the variance matrix of the pseudorange measurements,
- $\sigma_{\Phi,u,m}^{s} = m_{\Phi,u}^{s} \sigma_{\Phi,u,m}$ is the standard deviation of carrier phase measurements for the signal m from the satellite s generated by the receiver of the user u,
- $m_{\Phi,u}^{s} = 1/sin\left(\theta_u^s\right)$ is the mapping function for the variance of the carrier phase measurements,
- $\sigma_{\Phi,u,m}$ is the standard deviation of the carrier phase measurements for the signal m from a hypothetical satellite at zenith,

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If the full $\mathbf{\Sigma}_{cc}$ matrix is not available, it can be approximated as

$$\mathbf{\Sigma}_{\overline{cc}} = \mathrm{diag}\left(\mathbf{\Sigma}_{r^s}, \mathbf{\Sigma}_{v^s}, \mathbf{\Sigma}_{dt^s}, \mathbf{\Sigma}_{\dot{dt}^s}, \mathbf{\Sigma}_{\beta^s}, \mathbf{\Sigma}_{b^s}, \mathbf{\Sigma}_{T_H}, \mathbf{\Sigma}_W, \mathbf{\Sigma}_{\mathrm{PCO}}, \mathbf{\Sigma}_{\mathrm{PCV}}\right) \tag{105}$$

and $\mathbf{\Sigma}_{\tilde{\mathbf{z}}_k}$ can be instead computed as

$$\mathbf{\Sigma}_{\tilde{\mathbf{z}}_k} = \mathbf{T}_{\mathrm{SD}} \begin{bmatrix} \mathbf{\Sigma}_{\overline{yy}} & \mathbf{0} \\ \mathbf{0} & \alpha_u \mathbf{\Sigma}_{\overline{cc}} \end{bmatrix} \mathbf{T}_{\mathrm{SD}}{}^{T} \tag{106}$$

where

- $\mathbf{\Sigma}_{r^s} = \mathrm{diag}(\mathbf{\Sigma}_{r^E}, \mathbf{\Sigma}_{r^G})$, $\mathbf{\Sigma}_{r^\Sigma} = \bigoplus_{s=1}^{S_\Sigma} \bigoplus_{c=x,y,z} \sigma_{r_c^s}{}^2$,
- $\mathbf{\Sigma}_{v^s} = \mathrm{diag}(\mathbf{\Sigma}_{v^E}, \mathbf{\Sigma}_{v^G})$, $\mathbf{\Sigma}_{v^\Sigma} = \bigoplus_{s=1}^{S_\Sigma} \bigoplus_{c=x,y,z} \sigma_{v_c^s}{}^2$,
- $\sigma_{r_c^s}$ and $\sigma_{v_c^s}$, $c = x, y, z$ are the position and velocity accuracies of satellite s,
- $\mathbf{\Sigma}_{dt^s} = \mathrm{diag}(\mathbf{\Sigma}_{dt^E}, \mathbf{\Sigma}_{dt^G})$, $\mathbf{\Sigma}_{dt^\Sigma} = \bigoplus_{s=1}^{S_\Sigma} \sigma_{dt^s}{}^2$,
- $\mathbf{\Sigma}_{\dot{dt}^s} = \mathrm{diag}(\mathbf{\Sigma}_{\dot{dt}^E}, \mathbf{\Sigma}_{\dot{dt}^G})$, $\mathbf{\Sigma}_{\dot{dt}^\Sigma} = \bigoplus_{s=1}^{S_\Sigma} \sigma_{\dot{dt}^s}{}^2$,
- σ_{dt^s} and $\sigma_{\dot{dt}^s}$ are the clock error and drift accuracies of satellite s,
- $\mathbf{\Sigma}_{b^s} = \mathrm{diag}(\mathbf{\Sigma}_{b^E}, \mathbf{\Sigma}_{b^G})$, $\mathbf{\Sigma}_{b^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \sigma_{b_m^s}{}^2$,
- $\sigma_{b_m^s}$ is the code bias accuracy of satellite s for signal m,
- $\mathbf{\Sigma}_{T_\mathrm{h}} = \mathrm{diag}\left(\mathbf{\Sigma}_{T_h^E}, \mathbf{\Sigma}_{T_h^G}\right)$, $\mathbf{\Sigma}_{T_h^\Sigma} = \bigoplus_{s=1}^{S_\Sigma} \sigma_{T_h^s}{}^2$,
- $\sigma_{T_h^s}$ is the slant hydrostatic tropospheric error accuracy,
- $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^s} = \mathrm{diag}\left(\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^E}, \mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^G}\right)$, $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \sigma_{\tilde{\boldsymbol{\zeta}}_{PCO,u,m}^s}{}^2$
- $\sigma_{\tilde{\boldsymbol{\zeta}}_{PCO,u,m}^s}$ is the phase centre offset accuracy of satellite s and receiver u for signal m.$\mathbf{\Sigma}_{b^s} = \mathrm{diag}(\mathbf{\Sigma}_{b^E}, \mathbf{\Sigma}_{b^G})$, $\mathbf{\Sigma}_{b^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \left(\alpha_{u,m} \sigma_{b_m^s}{}^2\right)$
- $\mathbf{\Sigma}_{\beta^s} = \mathrm{diag}\left(\mathbf{\Sigma}_{\beta^E}, \mathbf{\Sigma}_{\beta^G}\right)$, $\mathbf{\Sigma}_{\beta^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \left(\alpha_{u,m} \sigma_{\beta_m^s}{}^2\right)$,
- $\sigma_{\beta_m^s}$ is the phase bias accuracy of satellite s for signal m,
- $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^s} = \mathrm{diag}\left(\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^E}, \mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^G}\right)$, $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCO,u}^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \left(\alpha_{u,m} \sigma_{\tilde{\boldsymbol{\zeta}}_{PCO,u,m}^s}{}^2\right)$
- $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCV,u}^s} = \mathrm{diag}\left(\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCV,u}^E}, \mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCV,u}^G}\right)$, $\mathbf{\Sigma}_{\tilde{\boldsymbol{\zeta}}_{PCV,u}^\Sigma} = \bigoplus_{m=1}^{M_\Sigma} \bigoplus_{s=1}^{S_\Sigma} \left(\alpha_{u,m} \sigma_{\tilde{\boldsymbol{\zeta}}_{PCV,u,m}^s}{}^2\right)$
- $\sigma_{\tilde{\boldsymbol{\zeta}}_{PCV,u,m}^s}$ is the phase centre variation accuracy of satellite s and receiver u for signal m,
- α_u is a user defined tuning parameter that compensates for unmodeled aspects of the corrections.

3.3.4. Design matrix determination

The design matrix is defined as in the following

$$\mathbf{H}_k = \begin{bmatrix} \mathbf{H}_{\Phi,r}^E & \mathbf{0} & \mathbf{0}_\mathrm{a} & \mathbf{H}_{\Phi,\delta T_w}^E & \mathbf{H}_{\Phi,I}^E & \mathbf{0} & \mathbf{H}_{\Phi,N}^E & \mathbf{0} \\ \mathbf{H}_{\Phi,r}^G & \mathbf{0} & \mathbf{0}_\mathrm{a} & \mathbf{H}_{\Phi,\delta T_w}^G & \mathbf{0} & \mathbf{H}_{\Phi,I}^G & \mathbf{0} & \mathbf{H}_{\Phi,N}^G \\ \mathbf{H}_{P,r}^E & \mathbf{0} & \mathbf{0}_\mathrm{a} & \mathbf{H}_{P,\delta T_w}^E & \mathbf{H}_{P,I}^E & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{H}_{P,r}^G & \mathbf{0} & \mathbf{0}_\mathrm{a} & \mathbf{H}_{P,\delta T_w}^G & \mathbf{0} & \mathbf{H}_{P,I}^G & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{D,v}^E & \mathbf{0}_\mathrm{a} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{D,v}^G & \mathbf{0}_\mathrm{a} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{107}$$

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where, for $\Sigma = E|G$,

$$\mathbf{H}_{\Phi,r}^{\Sigma} = \mathbf{H}_{P,r}^{\Sigma} = \mathbf{H}_{D,v}^{\Sigma} = \mathbf{1}_{M_\Sigma} \otimes \begin{bmatrix} \mathbf{e}_u^1 & \cdots & \mathbf{e}_u^{p_\Sigma - 1} & \mathbf{e}_u^{p_\Sigma + 1} & \cdots & \mathbf{e}_u^{S_\Sigma} \end{bmatrix}^T - \mathbf{1}_{S_\Sigma - 1} \otimes \mathbf{e}_u^{p_\Sigma} \tag{108}$$

$$\mathbf{H}_{\Phi,\delta T_w}^{\Sigma} = \mathbf{H}_{P,\delta T_w}^{\Sigma} = \begin{bmatrix} m_{\ni,\mathrm{u}}^1 & \cdots & m_{\ni,\mathrm{u}}^{p_\Sigma - 1} & m_{\ni,\mathrm{u}}^{p_\Sigma + 1} & \cdots & m_{\ni,\mathrm{u}}^{S_\Sigma} \end{bmatrix}^T - m_{\ni,\mathrm{u}}^{p_\Sigma} \mathbf{1}_{S_\Sigma - 1} \tag{109}$$

$$\mathbf{H}_{\Phi,I}^{\Sigma} = -\mathbf{H}_{\mathrm{P},I}^{\Sigma} = -\begin{bmatrix} \mu_1 \\ \vdots \\ \mu_{M_\Sigma} \end{bmatrix} \otimes \mathbf{I}_{S_\Sigma - 1} \tag{110}$$

$$\mathbf{H}_{\Phi,N}^{\Sigma} = \left[\bigoplus_{m=1}^{M_\Sigma} \lambda_m\right] \otimes \mathbf{I}_{S_\Sigma - 1} \tag{111}$$

3.3.5. Initialization of ionosphere and ambiguity states of newly tracked satellites

The SD ionosphere slant delays and the double difference (DD), if satellite phase biases are applied, SD otherwise, ambiguities of new tracked satellites are also initialized by applying a LS technique as in the following

$$\underline{\hat{\mathbf{x}}}_k^{\mathrm{new}} = \left(\mathbf{H}_2^{\mathrm{T}} \boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k,\mathrm{new}}^{-1} \mathbf{H}_2\right)^{-1} \mathbf{H}_2^{\mathrm{T}} \boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k,\mathrm{new}}^{-1} \left(\underline{\tilde{\mathbf{z}}}_k^{p,\mathrm{new}} - \mathbf{H}_1 \begin{bmatrix} \underline{\check{\mathbf{r}}}_k \\ \check{I}_{\delta \mathrm{w} k} \end{bmatrix}\right) \tag{112}$$

where

- $\underline{\hat{\mathbf{x}}}_k^{\mathrm{new}} = \begin{bmatrix} \hat{\mathbf{I}}_u^{s_\mathrm{E} p,\mathrm{new}T} & \hat{\boldsymbol{I}}_u^{s_\mathrm{G} p,\mathrm{new}T} & \hat{\boldsymbol{N}}_{u,1_E}^{s_\mathrm{E} p,\mathrm{new}T} & \cdots & \hat{\boldsymbol{N}}_{u,M_E}^{s_\mathrm{E} p,\mathrm{new}T} & \hat{\boldsymbol{N}}_{u,1_G}^{s_\mathrm{G} p,\mathrm{new}T} & \cdots & \hat{\boldsymbol{N}}_{u,M_G}^{s_\mathrm{G} p,\mathrm{new}T} \end{bmatrix}^T$,
- $\hat{\boldsymbol{I}}_u^{s_\Sigma p,\mathrm{new}} = \begin{bmatrix} I_u^{1_\Sigma p_\Sigma}, \ldots, I_u^{(S_\Sigma - 1) p_\Sigma} \end{bmatrix}^T$, for $\Sigma = \mathrm{E}|\mathrm{G}$,
- $\hat{\boldsymbol{N}}_{u,m}^{s_\Sigma p,\mathrm{new}} = \begin{bmatrix} N_{u,m}^{1_\Sigma p_\Sigma}, \ldots, N_{u,m}^{(S_\Sigma - 1) p_\Sigma} \end{bmatrix}^T$, for $\Sigma = \mathrm{E}|\mathrm{G}$ and $m = 1_\Sigma, \ldots, M_\Sigma$,
- $\underline{\tilde{\mathbf{z}}}_k^{p,\mathrm{new}}$ and $\boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k,\mathrm{new}}$ are the vectors of the observed minus computed SD measurements and their variance matrix as defined in Section 3.3.3 and computed for the measurements of the satellites flagged as "new".

The partial mapping matrices H_1 and H_2 are given by

$$\boldsymbol{H_1} = \begin{bmatrix} \mathbf{H}_{\Phi,r}^{E,\mathrm{new}} & \mathbf{H}_{\Phi,\delta T_w}^{E,\mathrm{new}} \\ \mathbf{H}_{\Phi,r}^{G,\mathrm{new}} & \mathbf{H}_{\Phi,\delta T_w}^{G,\mathrm{new}} \\ \mathbf{H}_{\mathrm{P},r}^{E,\mathrm{new}} & \mathbf{H}_{\mathrm{P},\delta T_w}^{E,\mathrm{new}} \\ \mathbf{H}_{\mathrm{P},r}^{G,\mathrm{new}} & \mathbf{H}_{\mathrm{P},\delta T_w}^{G,\mathrm{new}} \end{bmatrix} \quad \text{and} \quad \boldsymbol{H_2} = \begin{bmatrix} \mathbf{H}_{\Phi,I}^{E,\mathrm{new}} & \mathbf{0} & \mathbf{H}_{\Phi,N}^{E,\mathrm{new}} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{\mathrm{P},I}^{G,\mathrm{new}} & \mathbf{0} & \mathbf{H}_{\Phi,N}^{G,\mathrm{new}} \\ \mathbf{H}_{\mathrm{P},I}^{E,\mathrm{new}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_{\mathrm{P},I}^{G,\mathrm{new}} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{113}$$

where the design submatrices $\mathbf{H}_{\Phi,r}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\mathrm{P},r}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\Phi,\delta T_w}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\mathrm{P},\delta T_w}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\Phi,I}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\mathrm{P},I}^{\Sigma,\mathrm{new}}, \mathbf{H}_{\Phi,N}^{\Sigma,\mathrm{new}}$ are defined in Section 3.3.3 and computed for the measurements of the satellites flagged as "new". $\underline{\hat{\mathbf{x}}}_k^{\mathrm{new}}$ is then used to initialise the corresponding parameters in the predicted state $\underline{\check{\mathbf{x}}}_k$.

The variance matrix of the new initialised parameters is computed as

$$\boldsymbol{\Sigma}_{\underline{\hat{\mathbf{x}}}_k^{\mathrm{new}}} = \mathbf{H}_2^{\mathrm{T}} \boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k,\mathrm{new}}^{-1} \mathbf{H}_2 \tag{114}$$

and then used to initialise the variances and covariance of the corresponding parameters in the covariance matrix of the predicted state vector $\boldsymbol{\Sigma}_{\check{\mathbf{x}}_k}$.

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3.3.6. Ambiguities and ionosphere slant delay reinitialization due to phase discontinuity

The HAS message provides a phase discontinuity indicator (PDI), which is a counter incremented every time a phase discontinuity occurs for a specific satellite and signal. This counter is mapped into a binary flag at the decoding stage and therefore before the algorithm. A "0" is used to indicate a continuous phase and a "1" refers to a discontinuous phase bias correction. In the case of phase discontinuity, or when a CS is detected by the receiver or the CS detection algorithm described in Section 3.2.7, the corresponding ambiguities and the ionosphere slant delays are reinitialized by the LS estimation described in Section 3.3.5 using the predicted receiver position estimate and the residual wet tropospheric zenith delay.

3.3.7. Computation of EKF process noise covariance matrix

The process noise covariance matrix indicates the uncertainty of the state-space model and can be expressed as

$$\boldsymbol{\Sigma}_{\mathbf{w}} = \begin{bmatrix} \boldsymbol{\Sigma}_{\bar{\mathbf{r}}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \boldsymbol{\Sigma}_{\bar{\mathbf{v}}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \boldsymbol{\Sigma}_{\bar{\mathbf{a}}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{\overline{\delta T}_{\mathrm{W}}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{\Sigma}_{\bar{I}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \boldsymbol{\Sigma}_{\bar{N}} \end{bmatrix} \tag{115}$$

where the sub-matrices/variables filling the matrix diagonal are reported in Table 5.

Table 5: Process noise covariance matrices.

Covariance Type	Definition
Covariance of the position	$\boldsymbol{\Sigma}_{\mathbf{r}} = \begin{bmatrix} \sigma^2_{r_{xn}} & 0 & 0 \\ 0 & \sigma^2_{r_{ye}} & 0 \\ 0 & 0 & \sigma^2_{r_{zu}} \end{bmatrix}$
Covariance of the velocity	$\boldsymbol{\Sigma}_{\mathbf{v}} = \begin{bmatrix} \sigma^2_{v_{xn}} & 0 & 0 \\ 0 & \sigma^2_{v_{ye}} & 0 \\ 0 & 0 & \sigma^2_{v_{zu}} \end{bmatrix}$
Covariance of the acceleration	$\boldsymbol{\Sigma}_{a} = \begin{bmatrix} \sigma^2_{a_{xn}} & 0 & 0 \\ 0 & \sigma^2_{a_{ye}} & 0 \\ 0 & 0 & \sigma^2_{a_{zu}} \end{bmatrix}$
Covariance of the troposphere	$\sigma^2_{\bar{T}_{\delta \mathrm{W}}}$
Covariance of the ionosphere	$\boldsymbol{\Sigma}_{I} = \begin{pmatrix} \sigma_I^2 & 0 & \cdots & 0 & 0 \\ 0 & \sigma_I^2 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \sigma_I^2 & 0 \\ 0 & 0 & \cdots & 0 & \sigma_I^2 \end{pmatrix}$
Covariance of the ambiguity states	$\boldsymbol{\Sigma}_{N} = \mathbf{0}_{M(N+K-2)\times M(N-K-2)}$

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Since these uncertainties are expected to be velocity-dependent, an adaptive scheme is used, i.e. the uncertainties of the velocity, acceleration, troposphere and ionosphere states are adapted according to the estimated user velocity. Specifically, the velocity estimated through SE PVT, as described in Section 3.2.5, is used for adapting the process noise standard deviations. Two velocity thresholds are considered, which result in three sub-cases, as summarized in Table 6. The user velocity thresholds and the variance values were empirically determined. For the process noise of the acceleration, only two cases are considered. For a user speed of at most 0.1 m/s, a small process noise of 0.2 $m/\left(s^2\sqrt{s}\right)$ is selected for the acceleration. For any speed above 0.1 m/s, a process noise of 3 $m/\left(s^2\sqrt{s}\right)$ is set. For the process noise of the atmospheric parameters, also extreme atmospheric conditions are considered. Therefore, the user should be aware of this aspect and consider using smaller values if mild conditions occur, for example by diminishing the ionosphere process noise down to 0.005 $\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ in low ionosphere activity at middle latitudes.

Table 6: Adaptive setting of the process noise standard deviations.

Velocity Condition	$\|v\| > 100\,\frac{\mathrm{m}}{\mathrm{s}}$	$\|v\| > 0.1\,\frac{\mathrm{m}}{\mathrm{s}}$	$\|v\| \leq 0.1\,\frac{\mathrm{m}}{\mathrm{s}}$
Process Noise	$\sigma_{r,n/e/u} = 0.02\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_{v,n/e/u} = 0.10\,\frac{\mathrm{m}}{s\sqrt{\mathrm{s}}}$ $\sigma_{a,n/e/u} = 3.00\,\frac{\mathrm{m}}{\mathrm{s}^2\sqrt{\mathrm{s}}}$ $\sigma_{T_z} = 0.002\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_I = 0.1\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$	$\sigma_{r,n/e/u} = 0.02\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_{v,n/e/u} = 0.02\,\frac{\mathrm{m}}{s\sqrt{\mathrm{s}}}$ $\sigma_{a,n/e/u} = 3.00\,\frac{\mathrm{m}}{\mathrm{s}^2\sqrt{\mathrm{s}}}$ $\sigma_{T_z} = 0.0002\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_I = 0.1\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$	$\sigma_{r,n/e/u} = 0.02\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_{v,n/e/u} = 0.02\,\frac{\mathrm{m}}{s\sqrt{\mathrm{s}}}$ $\sigma_{a,n/e/u} = 0.20\,\frac{\mathrm{m}}{\mathrm{s}^2\sqrt{\mathrm{s}}}$ $\sigma_{T_z} = 2\cdot 10^{-5}\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$ $\sigma_I = 0.1\,\frac{\mathrm{m}}{\sqrt{\mathrm{s}}}$

3.3.8. Measurement update

The measurement update allows updating the estimation of the state vector through the knowledge of the new measurements and the prediction of the state vector from its estimation at the previous epoch.

Estimation:

$$\underline{\boldsymbol{\nu}}_k = \underline{\tilde{\mathbf{z}}}_k^{\mathrm{p}} - \mathbf{H}_k\underline{\breve{\mathbf{x}}}_k \tag{116}$$

$$\boldsymbol{\Sigma}_{\nu_k} = \mathbf{H}_k\boldsymbol{\Sigma}_{\bar{\mathbf{x}}_k}\mathbf{H}_k^T + \boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k^p} \tag{117}$$

$$\mathbf{K}_k = \boldsymbol{\Sigma}_{\bar{\mathbf{x}}_k}\mathbf{H}_k^T\boldsymbol{\Sigma}_{\nu_k}^{-1} \tag{118}$$

$$\underline{\hat{\mathbf{x}}}_k = \underline{\breve{\mathbf{x}}}_k + \mathbf{K}_k\underline{\boldsymbol{\nu}}_k \tag{119}$$

$$\boldsymbol{\Sigma}_{\hat{x}_k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\boldsymbol{\Sigma}_{\hat{\mathbf{x}}_k}(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k\boldsymbol{\Sigma}_{\tilde{\mathbf{z}}_k^p}\mathbf{K}_k^T \tag{120}$$

where

- $\underline{\boldsymbol{\nu}}_k$ is the innovation vector,
- $\boldsymbol{\Sigma}_{\nu_k}$ is the innovation covariance matrix,
- $\mathbf{K}_k$ is the KF gain, weighting the errors between the predicted and real measurements,
- $\underline{\hat{\mathbf{x}}}_k$ is the state vector estimated by the measurement update,
- $\boldsymbol{\Sigma}_{\hat{x}_k}$ is the estimated state vector covariance matrix.

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3.3.9. Outlier detection and removal

After the NAV prediction and correction steps are performed, a DIA residual analysis is executed on the innovation vector $\underline{\boldsymbol{\nu}}_k$ and its covariance matrix $\boldsymbol{\Sigma}_{\nu_k}$ [18] following the same algorithm described in Section 3.2.6 but with some difference. As already specified in Section 3.2.6, the measurements are assumed to have a normal distribution and their stochastic model is identical in all the hypotheses considered, which are

$$H_o: \underline{\boldsymbol{\nu}}_k \sim N\left(0, \boldsymbol{\Sigma}_{\nu_k}\right) \tag{121}$$

and

$$H_a: \underline{\boldsymbol{\nu}}_k \sim N\left(\mathbf{C}_k \boldsymbol{\nabla}, \boldsymbol{\Sigma}_{\nu_k}\right) \tag{122}$$

In the null hypothesis it is assumed that the predicted residuals $\underline{\boldsymbol{\nu}}_k$ are normally distributed (denoted using $N(\cdot,\cdot)$) with zero mean and variance-covariance matrix $\boldsymbol{\Sigma}_{\nu_k}$, whereas in the alternative hypotheses it is assumed that they have a nonzero mean, denoted as $\mathbf{C}_k \boldsymbol{\nabla}$, with $\boldsymbol{\nabla}$ a q-vector of additional unknown parameters in the alternative hypotheses related to the expectation of $\underline{\boldsymbol{\nu}}_k$ by the $h \times q$ matrix $\mathbf{C}_k$ (also assumed full rank, $\mathrm{rank}(\mathbf{C}_\mathrm{k}) = \mathrm{q},\ 1 \leq q \leq h - l$) where q is the number of additional unknown parameters in the alternative hypothesis (also called degree of freedom), h is the total number of measurements and l is the number of parameters in the state vector without a dynamic constraint (in the NAV case $l = 0$).
The adopted method is based on the following steps.

1. ***Detection***. It is performed by adopting the estimator of the variance factor of unit weight, $\underline{\hat{\sigma}}^2$, where $q = h$, i.e. there is no redundancy in the alternative hypothesis H_a, implying that the predicted residuals of the alternative hypothesis are $\underline{\hat{\mathbf{e}}}_{u,a,i} = 0$. The expression of the $\underline{\hat{\sigma}}^2$-test statistic is

$$\underline{\hat{\sigma}}^2 = \frac{\underline{T}_{q=h-l}}{h} = \frac{\left\|\underline{\boldsymbol{\nu}}_k\right\|^2_{\boldsymbol{\Sigma}^{-1}_{\nu_k}}}{h} \sim F(h, \infty, \lambda) \tag{123}$$

   where $\underline{T}_{q=h} \sim \chi^2(h, \lambda)$ is the overall model test with non-centrality parameter $\lambda = 0$ for H_0. The null hypothesis H_0, corresponding to the nominal model described in Section 3.3.8, is rejected if

$$\underline{\hat{\sigma}}^2 > k^F_{\alpha_\sigma} = \frac{k^{\chi^2}_{\alpha_\sigma}}{h} \tag{124}$$

   where $k^F_{\alpha_\sigma}$ is the critical value for the F-distribution once that the probability of false alarm α_σ has been fixed. For numerical reasons it is more convenient to use the χ^2-distribution for the computation of the critical value using the relation in Eq. (49) (please note that this equality is valid only for the critical values, not for the distributions in general).

2. Identification. If the $\underline{\hat{\sigma}}^2$ test rejects the nominal model H_0, then it is necessary to identify where the error lies. A wide range of alternative hypotheses is considered since, at each time, an outlier is assumed to be present in just one of the observations and this is done for each observation. All the alternative hypotheses extend the null hypothesis with just a single ($q = 1$) additional unknown parameter ∇, so that $\mathbf{C}_k = \mathbf{c}_j = \left[\underbrace{0, \ldots, 0}_{j-1},\ 1, \underbrace{0, \ldots, 0}_{h-j},\right]^T$ and

$$H_{a_j}: E\left(\underline{\tilde{\mathbf{z}}}^{\mathrm{p}}_k\right) = \mathbf{H}_k \underline{\tilde{\mathbf{x}}}_k + \mathbf{c}_j \boldsymbol{\nabla}, \text{ for } j = 1, \ldots, h \tag{125}$$

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3. The h alternative hypotheses are tested using the w-test

$$\underline{w}_j = \frac{\mathbf{c}_j^T \boldsymbol{\Sigma}_{\nu_k}^{-1} \underline{\boldsymbol{\nu}}_k}{\sqrt{\mathbf{c}_j^T \boldsymbol{\Sigma}_{\nu_k}^{-1} \mathbf{c}_j}} \tag{126}$$

4. which for H_0 has distribution $N(1,0)$. The maximum is taken over the h different hypotheses $|w_a|$, for $a = 1, \dots, h$, with

$$\underset{j=1,\dots,h}{max} |w_j| = |w_k| \tag{127}$$

verifying the condition

$$|w_k| > \sqrt{k_{\frac{\alpha_w}{2}}^N}, \text{ where } \sqrt{k_{\frac{\alpha_w}{2}}^N} = \sqrt{k_{\alpha_w}^{\chi^2}} \tag{128}$$

The model error is assigned to the observation having the largest outlier test statistic exceeding the critical value.

5. Adaptation. Finally, the adaptation accounts for the identified error to avoid a biased solution accepting the null hypothesis. Hence, the identified alternative hypothesis H_{a_j} will become the new null hypothesis H_o computing, after calling $\mathbf{K}_k = \left(\mathbf{H}_k^T \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1} \mathbf{H}_k\right)^{-1} \mathbf{H}_k^T \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1}$,

$$\boldsymbol{\Sigma}_{\hat{\nabla}\hat{\nabla}} = \left(\mathbf{C}_k^T \boldsymbol{\Sigma}_{\nu_k}^{-1} \mathbf{C}_k\right)^{-1} \tag{129}$$

$$\underline{\hat{\boldsymbol{\nabla}}}_j = \boldsymbol{\Sigma}_{\hat{\nabla}\hat{\nabla}} \mathbf{C}_k^T \boldsymbol{\Sigma}_{\nu_k}^{-1} \underline{\boldsymbol{\nu}}_k \tag{130}$$

$$\underline{\hat{\mathbf{x}}}_{k,a} = \underline{\hat{\mathbf{x}}}_k - \mathbf{K}\mathbf{C}_k \underline{\hat{\boldsymbol{\nabla}}}_j = \underline{\hat{\mathbf{x}}}_k - \left(\mathbf{H}_k^T \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1} \mathbf{H}_k\right)^{-1} \mathbf{H}_k^T \boldsymbol{\Sigma}_{\tilde{y}\tilde{y}}^{-1} \mathbf{C}_k \underline{\hat{\boldsymbol{\nabla}}}_j \tag{131}$$

$$\boldsymbol{\Sigma}_{\hat{x}_a} = \boldsymbol{\Sigma}_{\hat{x}} + \mathbf{K}_k \mathbf{C}_k \boldsymbol{\Sigma}_{\nabla\nabla} \mathbf{c}_k^T \mathbf{K}_k^T \tag{132}$$

$$\boldsymbol{\lambda} = \underline{\hat{\boldsymbol{\nabla}}}_j^T \boldsymbol{\Sigma}_{\hat{\nabla}\hat{\nabla}}^{-1} \underline{\hat{\boldsymbol{\nabla}}}_j \tag{133}$$

As the data set may contain multiple errors, after a cycle of DIA, the procedure is to be started over, now with the new null hypothesis, until the Overall Model test in the detection step is no longer rejected, or until there is no redundancy. The same considerations and logic provided at the end of Section 3.2.6 are applicable also here.

3.3.10. Site displacement correction

The receiver position is affected by various displacement effects: solid Earth tides, polar motion and ocean loading. These effects can be compensated for an International Terrestrial Reference Frame (ITRF) compatible receiver position estimate. The mathematical description of the displacement effects and how these effects can be compensated is described in [6]. Please note that no Earth Rotation Parameters are provided by Galileo HAS in the Initial Service, therefore it is user's responsibility to procure the parameters for the site displacement computation.

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3.3.11. NAV estimation output

For completeness, the outputs of the NAV estimation block, introduced in Section 3.2, are recalled here:

- position estimates in GTRF;
- velocity estimates in GTRF;
- timing information in GST as provided by the PP block;
- 1-sigma accuracy of position estimates obtained from the covariance matrix of the state updates, provided in Section 3.3.8;
- 1-sigma accuracy of velocity estimates obtained from the covariance matrix of the state updates, provided in Section 3.3.8;
- 1-sigma accuracy of timing estimates obtained from the covariance matrix of the least-squares estimates, provided in Section 3.2.5.1.

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5. LIST OF ABBREVIATIONS

ANTEX	ANTenna EXchange Format
APC	Antenna Phase Centre
ARP	Antenna Reference Point
BC	BroadCast
BLUE	Best Linear Unbiased Estimator
C/N0	Carrier-to-Noise-Density Ratio
COM	Centre of Mass
CS	Cycle Slip
DIA	Detection Identification Adaptation
EC	European Commission
ECEF	Earth Centred Earth Fixed
ECI	Earth Centred Inertial
EKF	Extended Kalman Filter
ENU	East North Up
ERP	Earth Rotation Parameters
ESA	European Space Agency
EUSPA	EU Agency for the Space Program
EU	European Union
GF	Geometry Free
GAST	Greenwich Apparent Sidereal Time
GLR	Generalised Likelihood Ratio
GMF	Global Mapping Function
GMST	Greenwich Mean Sidereal Time
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPST	GPS Time
GST	Galileo System Time
GTRF	Galileo Terrestrial Reference Frame
HAS	High Accuracy Service
HAS-UA	High Accuracy Service User Algorithm
ICD	Interface Control Document
IDD	Internet Data Distribution
IERS	International Earth Rotation and Reference Systems Service

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IF	Ionosphere Free
ITRF	International Terrestrial Reference Frame
JRC	Joint Research Centre of EC
KF	Kalman Filter
LOM	Local Overall Model
LOS	Line of Sight
LS	Least Square
MOPS	Minimum Operational Performance Standard
MPL	Minimum Performance Level
MW	Melbourne-Wübbena
NAV	Navigation
NTRIP	Networked Transport of RTCM via Internet Protocol
NTW	Near-radial In-track Cross-track
ORO	Orbit Reference Offset
PCO	Phase Centre Offset
PCV	Phase Centre Variation
PDI	Phase Discontinuity Indicator
PP	Pre-Processing
PPP	Precise Point Positioning
PVT	Position Velocity Time
RTCA	Radio Technical Commission for Aeronautics
RTCM	Radio Technical Commission for Maritime Services
SBAS	Satellite Based Augmentation System
SCS	Satellite Centred System
SD	Single Difference
SDD	Service Definition Document
SE	Single Epoch
SIS	Signal In Space
STD	Standard Deviation
WL	Wide Lane
WLS	Weighted Least Square
WLSE	Weighted Least Square Estimation

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APPENDIX A – ANTENNA PHASE CENTRE VARIATION COMPUTATION

Scalar corrections for the phase-centre variation of the transmitting antennas $\zeta^{s}_{\mathrm{PCV,m}}$ and the receiving antenna $\zeta_{\mathrm{PCV,u,m}}$ are applied, which may be frequency, elevation, and azimuth dependent. The PCV values for the GNSS antennas are obtained by means of the antenna exchange format (ANTEX).

If azimuth dependent PCV values are provided, a bilinear interpolation over the azimuth and elevation dependant PCV values provided by the ANTEX is performed, otherwise a sole linear interpolation of the PCV values as function of the elevation is performed. The format of the PCV values is provided in [20].

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APPENDIX B – NOTATION DEFINITION

In this Appendix, the notation used for the equations included in the document is reported.

- Vectors (to be identified as column) are indicated with boldface, lower case characters, e.g. $\mathbf{x}$.
- $\|\cdot\|$ is used to indicate the Euclidean norm of a vector.
- $\|\cdot\|_{\boldsymbol{\Sigma}}$ is used to indicate the sigma-weighted norm of a vector.
- Matrices are indicated with boldface, upper case characters, e.g. $\mathbf{H}$.
- A horizontal underbar means that the variable indicates a stochastic process, e.g. $\underline{\mathbf{x}}$.
- A tilde means a priori "correction" (from model or estimated by the system) or "corrected", e.g. $\tilde{\varphi}$ indicates φ after application of corrections.
- A horizontal bar indicates the "estimable" parameter, e.g. $\bar{\mathbf{z}}$ is a combination of the original $\mathbf{z}$ parameter with other linearly dependent original parameters making the system estimable.
- A hat means "estimated", e.g. $\hat{\mathbf{t}}$ means the "estimated" $\bar{\mathbf{t}}$.
- A breve accent indicates the "predicted" parameter, e.g. $\breve{\mathbf{x}}$ is the prediction of $\hat{\mathbf{x}}$ according to a linear function.
- Δ and δ symbols are used to indicate space or time differences or corrections depending on the context.
- $\oplus$ is the direct sum operator: $\mathbf{A} \oplus \mathbf{B} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}$. In general, the direct sum of n matrices/vectors is $\bigoplus_{i=1}^{n} \mathbf{A}_i = \mathrm{diag}(\mathbf{A}_1, \dots, \mathbf{A}_n)$. It can also be used with multiple nested indices as $\bigoplus_{i=1}^{m} \bigoplus_{j=1}^{s} \mathbf{A}_i^{\mathrm{j}} = \mathrm{diag}\big(\mathrm{diag}(\mathbf{A}_1^1, \dots, \mathbf{A}_1^s), \dots, \mathrm{diag}(\mathbf{A}_m^1, \dots, \mathbf{A}_m^s)\big)$.
- $\otimes$ is the Kronecker product. If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is a $p \times q$ matrix, then the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $pm \times qn$ block matrix: $\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} \mathrm{a}_{11}\mathbf{B} & \dots & \mathrm{a}_{1\mathrm{n}}\mathbf{B} \\ \vdots & \ddots & \vdots \\ \mathrm{a}_{\mathrm{m}1}\mathbf{B} & \dots & \mathrm{a}_{\mathrm{mn}}\mathbf{B} \end{bmatrix}$.
- $\mathbf{1}_a = \underbrace{[1 \dots 1]}_{a \text{ times}}^T$.
- $\mathbf{C}_{a,b} = \begin{bmatrix} \mathbf{0}_{a \times b-a} \\ \mathbf{I}_{b-a} \end{bmatrix}$.
- $\mathbf{I}_a = \mathrm{diag}\,\underbrace{(1 \dots 1)}_{a \text{ times}}$.
- The rotation matrices $\mathbf{R}_1(\alpha)$, $\mathbf{R}_2(\alpha)$ and $\mathbf{R}_3(\alpha)$ around the principal axes 1,2,3 are defined as

$$\mathbf{R}_1(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{bmatrix} \tag{B.1}$$

$$\mathbf{R}_2(\alpha) = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix} \tag{B.2}$$

$$\mathbf{R}_3(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{B.3}$$

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APPENDIX C – ROTATION MATRICES FOR THE APPLICATION OF ANTEX FILES PHASE CENTRE OFFSETS

The terms $\delta\mathbf{r}_{PCO,u}$and $\delta\mathbf{r}^{s}_{PCO}$ are used to indicate the receiver and satellite PCOs corrections, that are expressed in the ECEF system. These terms are obtained by multiplying the PCOs obtained from the ANTEX file by a rotation matrix.

In the receiver case, the rotation matrix is defined as

$$\mathbf{A}_u = (\mathbf{e}_{u,x} \quad \mathbf{e}_{u,y} \quad \mathbf{e}_{u,z}) \tag{C.1}$$

with the unit vectors $\mathbf{e}_{u,x}$, $\mathbf{e}_{u,y}$, $\mathbf{e}_{u,z}$ indicating the receiver's attitude.

The above vectors can be obtained by the satellite position and the receiver position determined by the standard PVT or HAS solutions. For the GNSS receiver antenna, the z-axis is aligned with the up (zenith) direction of the local East-North-Up (ENU) coordinate frame. The x and y axes are aligned with the East and North axes to obtain a right-hand (ENU) coordinate system. The unit vectors are defined as

$$\mathbf{e}_{u,x} = \begin{pmatrix} -\sin(\lambda_u) \\ \cos(\lambda_u) \\ 0 \end{pmatrix} \tag{C.2}$$

$$\mathbf{e}_{u,y} = \begin{pmatrix} -\cos(\lambda_u)\sin(\varphi_u) \\ -\sin(\lambda_u)\sin(\varphi_u) \\ \cos(\varphi_u) \end{pmatrix} \tag{C.3}$$

$$\mathbf{e}_{u,z} = \begin{pmatrix} \cos(\lambda_u)\cos(\varphi_u) \\ \sin(\lambda_u)\cos(\varphi_u) \\ \sin(\varphi_u) \end{pmatrix} \tag{C.4}$$

where λ_u and φ_u denote the longitude and latitude of the receiver's position as determined by the standard PVT or HAS solution.

In the satellite case, the rotation matrix is defined as

$$\mathbf{A}^s = (\mathbf{e}^s_x \quad \mathbf{e}^s_y \quad \mathbf{e}^s_z) \tag{C.5}$$

with $\mathbf{e}^s_x, \mathbf{e}^s_y, \mathbf{e}^s_z$ defining the satellite attitude

The z-axis $\mathbf{e}^s_z$ points from the COM towards the centre of Earth. The y-axis $\mathbf{e}^s_y$ is the rotation axis of the solar panel, which is facing towards the Sun and is perpendicular to the z-axis. The x-axis is defined so as to compose a right-hand coordinate frame

$$\mathbf{e}^s_x = \mathbf{e}^s_y \times \mathbf{e}^s_z$$

$$\mathbf{e}^s_y = \mathbf{e}^s_z \times \mathbf{e}^s_{\text{Sun}}$$

$$\mathbf{e}^s_z = -\frac{\mathbf{r}^s}{\|\mathbf{r}^s\|}$$

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The normalized Sun vector points from the satellite to the Sun and is given by:

$$\mathbf{e}_{\text{Sun}}^{s} = \frac{\mathbf{r}_{\text{Sun}} - \mathbf{r}^{s}}{\|\mathbf{r}_{\text{Sun}} - \mathbf{r}^{s}\|}$$

with $\mathbf{r}_{\text{Sun}}$ indicating the position of the Sun and $\mathbf{r}^{s}$ indicating the position of the satellite s. The position of the Sun can be computed as indicated in Appendix D.

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APPENDIX D – SUN POSITION COMPUTATION

This Appendix provides details on the computation of the Sun position. The position of the Sun is first computed in the Earth-Centred Inertial (ECI) frame as $\mathbf{r}_{\text{Sun}}^{\text{ECI}}$ and then transformed into the ECEF coordinate frame using the rotation matrix $\mathbf{R}_{\text{ECI}}^{\text{ECEF}}$ as in the following (from [19])

$$\mathbf{r}_{\text{Sun}}^{\text{ECEF}} = \mathbf{R}_{\text{ECI}}^{\text{ECEF}} \mathbf{r}_{\text{Sun}}^{\text{ECI}} \tag{D.1}$$

The rotation matrix from the ECI into the ECEF system is defined as:

$$\mathbf{R}_{\text{ECI}}^{\text{ECEF}} = \mathbf{R_2}(-x_p)\mathbf{R_1}(-y_p)\mathbf{R_3}(\theta_{\text{a}})\,\mathbf{N}\,\mathbf{P} \approx \mathbf{R}_3(\theta_{\text{a}})\,\mathbf{N}\,\mathbf{P} \tag{D.2}$$

where

- (x_p, y_p) indicates the polar motion,
- θ_{a} indicates the Greenwich apparent sidereal time (GAST) expressed in radians,
- $\mathbf{N}$ indicates the nutation rotation matrix,
- $\mathbf{P}$ indicates the precession rotation matrix.

The polar motion can be neglected since it is too small to affect the PPP implementation. Therefore, the related term will not be considered in the following.

The GAST is computed by applying the following expression

$$\theta_{\text{a}} = \theta_m + \Delta\psi \cos\varepsilon \,(0.00264 \sin\alpha_5 + 0.000063 \sin(2\alpha_5))\frac{\pi}{180}\frac{1}{3600} \tag{D.3}$$

where

- θ_m is the Greenwich mean sidereal time (GMST) computed as indicated in Eq. (D.4),
- α_5 indicates the astronomical argument Ω, mean longitude of the ascending node of the Moon, computed as indicated in Eq. (D.8),
- $\Delta\psi$ is the nutation in longitude, whose computation is provided by Eq. (D.9),
- ε is the obliquity, whose computation is performed according to Eq. (D.7).

The GMST can be computed as

$$\theta_m = (\theta'_m / 86400)\frac{\pi}{43200} \tag{D.4}$$

with

$$\begin{aligned}\theta'_m = {} & 24110.54841 + 8640184.812866\, t_0 + 0.093104\, t_0^2 - 6.2 \cdot 10^{-6}\, t_0^3 \\ & + (1.002737909350795 + 5.9006 \cdot 10^{-11}\, t_0 - 5.9 \cdot 10^{-15}\, t_0^2)\, s_{\text{UT}}\end{aligned}$$

where $t_0 = d_0/36525$, d_0 being the number of days elapsed since J2000.0 (12h 1st January 2000) in UT1 (taking on values $\pm 0.5, \pm 1.5, \ldots$), and s_{UT} indicates the seconds of day in UT1 time.

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The nutation rotation matrix is defined as

$$\mathbf{N} = \mathbf{R_1}(-\varepsilon - \Delta\varepsilon)\mathbf{R}_3(-\Delta\psi)\,\mathbf{R}_1(\varepsilon) \tag{D.5}$$

with $\Delta\varepsilon$, the nutation in obliquity, defined in Eq. D.10.

The precession rotation matrix is defined as

$$\mathbf{P} = \mathbf{R_3}(-z)\mathbf{R_2}(\vartheta)\,\mathbf{R_3}(-\zeta) \tag{D.6}$$

where the parameters ζ, ϑ and z are computed, together with the previously used ε, by applying Eq. (D.7)

$$\begin{pmatrix}\zeta\\ \vartheta\\ z\\ \varepsilon\end{pmatrix} = \begin{pmatrix}2306.2181 & 0.30188 & 0.0179980\\ 2004.3109 & -0.42665 & -0.041833\\ 2306.2181 & 1.09468 & 0.018203\\ -46.8150 & -0.00059 & 0.001813\end{pmatrix}\begin{pmatrix}t_{\mathrm{TT}}\\ t_{\mathrm{TT}}^2\\ t_{\mathrm{TT}}^3\end{pmatrix}\frac{\pi}{180}\,\frac{1}{3600} \tag{D.7}$$

with t_{TT} Terrestrial Time in Julian centuries since J2000.0.

The astronomical arguments are given by

$$\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\\ \alpha_4\\ \alpha_5\end{pmatrix} = \begin{pmatrix}134.96340251\\ 357.52910918\\ 93.27209062\\ 297.85019547\\ 125.04455501\end{pmatrix} + \frac{1}{3600}\begin{pmatrix}1717915923.2178 & 31.8792 & 0.051635 & -0.00024470\\ 129596581.0481 & -0.5532 & 0.000136 & -0.00001149\\ 1739527262.8478 & -12.7512 & -0.001037 & 0.00000417\\ 1602961601.2090 & -6.3706 & 0.006593 & -0.00003169\\ -6962890.2665 & 7.4722 & 0.007702 & -0.00005939\end{pmatrix}\begin{pmatrix}t_{TT}\\ t_{TT}^2\\ t_{\mathrm{TT}}^3\\ t_{\mathrm{TT}}^4\end{pmatrix} \tag{D.8}$$

The geodesic nutation in longitude $\Delta\psi$ and obliquity $\Delta\varepsilon$ are approximated by a linear combination of 106 trigonometric functions,

$$\Delta\psi = \sum_{i=1}^{106}(a_{0,i} + a_{1,i}\,t_{TT})\sin(\omega_i) \tag{D.9}$$

$$\Delta\varepsilon = \sum_{i=1}^{106}(b_{0,i} + b_{1,i}t_{TT})\cos(\omega_i) \tag{D.10}$$

with numerically determined coefficients $a_{0,i}, a_{1,i}, b_{0,i}, b_{1,i}$, defined in Table D.1 and

$$\omega_i = \sum_{j=1}^{5} c_{j,i}\,\alpha_j \tag{D.11}$$

with the astronomical arguments α_j, provided by Eq. (D.8), and the coefficients $c_{j,i}$ given in Table D.1.

Table D.1: Coefficients for the computation of the terms $\boldsymbol{\Delta\psi}$, $\boldsymbol{\Delta\varepsilon}$, and $\boldsymbol{\omega_i}$.

i	$c_{1,i}$	$c_{2,i}$	$c_{3,i}$	$c_{4,i}$	$c_{5,i}$	$a_{0,i}$	$a_{1,i}$	$b_{0,i}$	$b_{1,i}$
1	0	0	0	0	1	-171996	-174.2	92025	8.9
2	0	0	2	-2	2	-13187	-1.6	5736	-3.1
3	0	0	2	0	2	-2274	-0.2	977	-0.5
4	0	0	0	0	2	2062	0.2	-895	0.5

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5	0	-1	0	0	0	-1426	3.4	54	-0.1
6	1	0	0	0	0	712	0.1	-7	0
7	0	1	2	-2	2	-517	1.2	224	-0.6
8	0	0	2	0	1	-386	-0.4	200	0
9	1	0	2	0	2	-301	0	129	-0.1
10	0	-1	2	-2	2	217	-0.5	-95	0.3
11	-1	0		2	0	158	0	-1	0
12	0	0	2	-2	1	129	0.1	-70	0
13	-1	0	2	0	2	123	0	-53	0
14	1	0	0	0	1	63	0.1	-33	0
15	0	0	0	2	0	63	0	-2	0
16	-1	0	2	2	2	-59	0	26	0
17	-1	0	0	0	1	-58	-0.1	32	0
18	1	0	2	0	1	-51	0	27	0
19	-2	0	0	2	0	-48	0	1	0
20	-2	0	2	0	1	46	0	-24	0
21	0	0	2	2	2	-38	0	16	0
22	2	0	2	0	2	-31	0	13	0
23	2	0	0	0	0	29	0	-1	0
24	1	0	2	-2	2	29	0	-12	0
25	0	0	2	0	0	26	0	-1	0
26	0	0	2	-2	0	-22	0	0	0
27	-1	0	2	0	1	21	0	-10	0
28	0	2	0	0	0	17	-0.1	0	0
29	0	2	2	-2	2	-16	0.1	7	0
30	-1	0	0	2	1	16	0	-8	0
31	0	1	0	0	1	-15	0	9	0
32	1	0	0	-2	1	-13	0	7	0
33	0	-1	0	0	1	-12	0	6	0
34	2	0	-2	0	0	11	0	0	0
35	-1	0	2	2	1	-10	0	5	0
36	1	0	2	2	2	-8	0	3	0
37	0	-1	2	0	2	-7	0	3	0
38	0	0	2	2	1	-7	0	3	0
39	1	1	0	-2	0	-7	0	0	0

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40	0	1	2	0	2	7	0	-3	0
41	-2	0	0	2	1	-6	0	3	0
42	0	0	0	2	1	-6	0	3	0
43	2	0	2	-2	2	6	0	-3	0
44	1	0	0	2	0	6	0	0	0
45	1	0	2	-2	1	6	0	-3	0
46	0	0	0	-2	1	-5	0	3	0
47	0	-1	2	-2	1	-5	0	3	0
48	2	0	2	0	1	-5	0	3	0
49	1	-1	0	0	0	5	0	0	0
50	1	0	0	-1	0	-4	0	0	0
51	0	0	0	1	0	-4	0	0	0
52	0	1	0	-2	0	-4	0	0	0
53	1	0	-2	0	0	4	0	0	0
54	2	0	0	-2	1	4	0	-2	0
55	0	1	2	-2	1	4	0	-2	0
56	1	1	0	0	0	-3	0	0	0
57	1	-1	0	-1	0	-3	0	0	0
58	-1	-1	2	2	2	-3	0	1	0
59	0	-1	2	2	2	-3	0	1	0
60	1	-1	2	0	2	-3	0	1	0
61	3	0	2	0	2	-3	0	1	0
62	-2	0	2	0	2	-3	0	1	0
63	1	0	2	0	0	3	0	0	0
64	-1	0	2	4	2	-2	0	1	0
65	1	0	0	0	2	-2	0	1	0
66	-1	0	2	-2	1	-2	0	1	0
67	0	-2	2	-2	1	-2	0	1	0
68	-2	0	0	0	1	-2	0	1	0
69	2	0	0	0	1	2	0	-1	0
70	3	0	0	0	0	2	0	0	0
71	1	1	2	0	2	2	0	-1	0
72	0	0	2	1	2	2	0	-1	0
73	1	0	0	2	1	-1	0	0	0
74	1	0	2	2	1	-1	0	1	0

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75	1	1	0	-2	1	-1	0	0	0
76	0	1	0	2	0	-1	0	0	0
77	0	1	2	-2	0	-1	0	0	0
78	0	1	-2	2	0	-1	0	0	0
79	1	0	-2	2	0	-1	0	0	0
80	1	0	-2	-2	0	-1	0	0	0
81	1	0	2	-2	0	-1	0	0	0
82	1	0	0	-4	0	-1	0	0	0
83	2	0	0	-4	0	-1	0	0	0
84	0	0	2	4	2	-1	0	0	0
85	0	0	2	-1	2	-1	0	0	0
86	-2	0	2	4	2	-1	0	1	0
87	2	0	2	2	2	-1	0	0	0
88	0	-1	2	0	1	-1	0	0	0
89	0	0	-2	0	1	-1	0	0	0
90	0	0	4	-2	2	1	0	0	0
91	0	1	0	0	2	1	0	0	0
92	1	1	2	-2	2	1	0	-1	0
93	3	0	2	-2	2	1	0	0	0
94	-2	0	2	2	2	1	0	-1	0
95	-1	0	0	0	2	1	0	-1	0
96	0	0	-2	2	1	1	0	0	0
97	0	1	2	0	1	1	0	0	0
98	-1	0	4	0	2	1	0	0	0
99	2	1	0	-2	0	1	0	0	0
100	2	0	0	2	0	1	0	0	0
101	2	0	2	-2	1	1	0	-1	0
102	2	0	-2	0	1	1	0	0	0
103	1	-1	0	-2	0	1	0	0	0
104	-1	0	0	1	1	1	0	0	0
105	-1	-1	0	2	1	1	0	0	0
106	0	1	0	1	0	1	0	0	0

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The position of the Sun in the ECI frame is given by [21]

$$\mathbf{r}_{\mathrm{Sun}} = \mathrm{r}_{\mathrm{Sun}} \begin{pmatrix} \cos\left(\frac{\pi}{180} l_{\mathrm{Sun}}\right) \\ \cos(e)\sin\left(\frac{\pi}{180} l_{\mathrm{Sun}}\right) \\ \sin(e)\sin\left(\frac{\pi}{180} l_{\mathrm{Sun}}\right) \end{pmatrix} \tag{D.12}$$

where r_{Sun} is the distance of the Sun computed as

$$r_{\mathrm{Sun}} = \mathrm{AU}\left(1.000140612 - 0.016708617 \cdot \cos\left(\frac{\pi}{180} M_{\mathrm{Sun}}\right) - 0.000139589 \cdot \cos\left(\frac{2\pi}{180} M_{\mathrm{Sun}}\right)\right) \tag{D.13}$$

with AU indicating the Astronomical Unit (AU = 149597870700 m).

The ecliptic longitude of the Sun l_{Sun} is given by (in degrees)

$$l_{\mathrm{Sun}} = 280.460 + 36000.770\, t_{\mathrm{TT}} + 1.914666471 \cdot \sin\left(\frac{\pi}{180} M_{\mathrm{Sun}}\right) + 0.019994643 \cdot \sin\left(\frac{2\pi}{180} M_{\mathrm{Sun}}\right) \tag{D.14}$$

The mean obliquity e of the ecliptic is given by (in degrees)

$$e = 23.439291 - 0.0130042\, \mathrm{t}_{\mathrm{TT}} \tag{D.15}$$

The mean anomaly of the Sun M_{Sun} is given by (in degrees)

$$M_{\mathrm{Sun}} = 357.5277233 + 35999.05034\, t_{\mathrm{TT}} \tag{D.16}$$

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APPENDIX E – PHASE WIND-UP EFFECT COMPUTATION

GNSS signals are right-hand polarized signals. The rotation of the satellite s and receiver u antennae around their boresight axes leads to a shift in the carrier phase indicated as the phase wind-up effect.

For PPP implementation, this phase wind-up effect needs to be considered and requires the application of correction $\lambda_m W^s_{\mathrm{u,m}}$ that can be determined as

$$W^s_{\mathrm{u,m}}(t_i) = \delta w^s_{u,m}(t_i) + 2N\pi = \delta w^s_{u,m}(t_i) + 2\pi\,\mathrm{nint}\left[\frac{W^s_{\mathrm{u,m}}(t_{i-1}) - \delta w^s_{u,m}(t_i)}{2\pi}\right] \tag{E.1}$$

$$\delta w^s_{u,m}(t_i) = \mathrm{sign}\left(\mathbf{e}^{s\,T}_u(\mathbf{d}^s \times \mathbf{d}_u)\right)\arccos\left(\frac{\mathbf{d}^s \cdot \mathbf{d}_u}{\|\mathbf{d}^s\|\|\mathbf{d}_u\|}\right) \tag{E.2}$$

where N is an integer number that can be initialised as zero, $W^s_{\mathrm{u,m}}(t_{i-1})$ is the previous value of the phase wind-up correction, $\mathbf{d}_u$ and $\mathbf{d}^s$ indicate the dipoles of the antennas of the user u and the satellite s and are given according to [16] by

$$\mathbf{d}_u = \mathbf{e}_{x,u} - \mathbf{e}^s_u(\mathbf{e}^s_u \cdot \mathbf{e}_{x,u}) + \mathbf{e}^s_u \times \mathbf{e}_{y,u} \tag{E.3}$$

$$\mathbf{d}^s = \mathbf{e}^s_x - \mathbf{e}^s_u(\mathbf{e}^s_u \cdot \mathbf{e}^s_x) - \mathbf{e}^s_u \times \mathbf{e}^s_y \tag{E.4}$$

with the unit vectors $\mathbf{e}^s_x, \mathbf{e}^s_y, \mathbf{e}^s_z$, provided in Appendix C, defining the attitude of the satellite, the unit vectors $\mathbf{e}_{x,u}, \mathbf{e}_{y,u}, \mathbf{e}_{z,u}$ defining the receiver's attitude, and $\mathbf{e}^s_u$ indicating the unit vector from the satellite to the receiver.

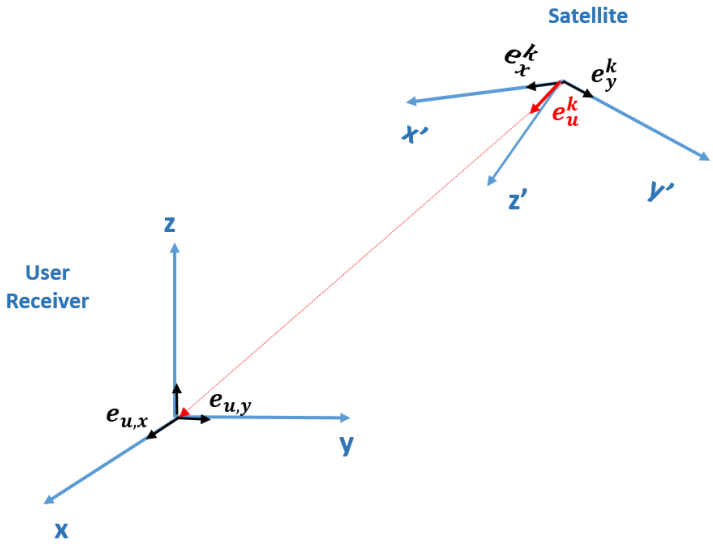


Figure E.1: User/satellite dipoles representation.

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APPENDIX F – NUMERICAL VALIDATION OF THE HAS UA

This appendix provides the numerical values for the HAS UA parameters adopted for its validation. Parameters marked with an asterisk (*) are GNSS receiver-specific; the user must characterize them specifically for the adopted hardware. Measurements with σ_P values larger than 1 meter (m) are not adequate for use in these algorithms to achieve the HAS performance outlined in the SDD.

Parameter	Symbol	Value
Elevation cut-off threshold	tr_{co}	5°
C/N0 threshold	tr_{cn0}	20 dB-Hz
WLS maximum number of iterations	$\mathrm{tr}_{\mathrm{WLS}_i}$	25
WLS residual convergence threshold	$\mathrm{tr}_{\delta\hat{\Sigma}_{u,i}}$	10^{-12} m
Probability of false alarm for LOM-test	α_{LOM}	0.05
Probability of false alarm for w-test	α_w	0.001
Pseudorange noise Normal STD E1/L1/L2	$\sigma_{P,u,E1/L1/L2}$	Receiver dependent * (e.g. 0.2 m)
Pseudorange noise Normal STD E5a/E5b/E6	$\sigma_{P,u,\mathrm{E5a/E5b/E6}}$	Receiver dependent * (e.g. 0.1 m)
Carrier phase noise Normal STD (all)	$\sigma_{\Phi,u,m}$	Receiver dependent * (e.g. 0.002 m)
Doppler noise Normal STD (all)	$\sigma_{D,u,\mathrm{m}}$	Receiver dependent * (e.g. 0.2 m/s)
Satellite position accuracy Normal STD (NTW)	$\mathbf{\Sigma}_{r^s}$	0.1, 0.1, 0.1 (GAL-GPS) m
Satellite velocity accuracy Normal STD (NTW)	$\mathbf{\Sigma}_{v^s}$	0.01, 0.01, 0.01 (GAL-GPS) m/s
Satellite clock error accuracy Normal STD	$\mathbf{\Sigma}_{dt^s}$	0.2 (GAL-GPS) m
Satellite clock drift accuracy Normal STD	$\mathbf{\Sigma}_{\dot{dt}^s}$	0.02 (GAL-GPS) m/s
Satellite code bias accuracy Normal STD	$\mathbf{\Sigma}_{d^s}$	0.02 (E1C/L1CA/E5aQ/E5bQ/E6B(C)/L2W/L2C) m
Melbourne-Wübbena combination CS threshold	k_{mw}	2
Sample number for Melbourne-Wübbena CS	l_{mw}	8
Sample number for geometry-free CS	l_{gf}	8
Ionosphere rate CS maximum threshold	$a_{0,\mathrm{ir}}$	0.1 m
Ionosphere rate CS factor ratio	α_{ir}	0.5

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Parameter	Symbol	Value
Geometry-free CS maximum threshold	$a_{0,\mathrm{gf}}$	0.03 m
Geometry-free CS factor ratio	α_{gf}	0.5
CS maximum tolerable gap	T	60 s
Doppler integration CS thresholds	$k_{\mathrm{di}}, f_{\mathrm{di}}$	0.9, 0.1 cyc
Tuning parameter for unmodeled correction	α_u	1/100

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